

Exercises**Level - 1****(Practice Questions Based on Fundamentals)****ABC OF PROBABILITY**

1. A coin is tossed twice. Find the probability of getting
 - (i) two heads
 - (ii) exactly one head
 - (iii) exactly two tails

2. A coin is tossed three times. Find the probability of getting
 - (i) exactly 2 heads
 - (ii) at least 2 heads
 - (iii) exactly 1 head.

3. A die is thrown twice. Find the probability of getting
 - (i) a sum of 10.
 - (ii) a sum of at least 9.
 - (iii) a sum of even numbers
 - (iv) a sum of odd numbers
 - (v) a sum of perfect numbers
 - (vi) both the dice show prime numbers.

4. A die is thrown 3 times. Find the probability of getting
 - (i) a sum of 3
 - (ii) a sum of 4
 - (iii) a sum of 5
 - (iv) a sum of 6.
 - (v) a sum of 7
 - (vi) a sum of at least 15
 - (vii) a sum of at most 16.

5. A card is drawn from a pack of 52 cards. Find the probability of getting
 - (i) a red card

- (ii) a black card
 - (iv) a king of a red colour
 - (v) a queen of spade
 - (vi) a face card.
6. Two cards are drawn at random from a well-shuffled pack of 52 cards. Find the probability of getting
- (i) both are red cards
 - (ii) both are kings
 - (iii) one is face card and another one is a king of a red colour.
 - (iv) one is court card and another one is an ace card
 - (v) both are of the same suit
 - (vi) both are of the different suit.
7. Four cards are drawn at random from a pack of 52 cards. Find the probability of getting
- (i) 2 red cards and 2 kings of black colour
 - (ii) all are of different suit
 - (iii) all are of same suit
 - (iv) all are honours cards
 - (v) all are face cards
 - (vi) 3 are court cards
 - (vii) 2 are aces
 - (viii) 2 slave cards
 - (ix) all are number cards
 - (x) all are same number cards.
8. A natural number x is chosen at random from first 100 natural numbers. Find the probability of getting
- (i) an even number
 - (ii) an odd number
 - (iii) a prime number
 - (iv) a perfect number
 - (v) a number which have only three factors
 - (vi) a number which have only four factors
 - (vii) a perfect square
 - (viii) a perfect cube
 - (ix) a perfect fourth power
 - (x) a perfect sixth power.
9. Two natural numbers are drawn at random from 100 natural numbers. Find the probability that
- (i) both are divisible by 2 and 3
 - (ii) both are divisible by 3 and 4
 - (iii) both are divisible by 3 and 7
 - (iv) both are divisible by 3 and 5
 - (v) one is divisible by 2 and another one is divisible by 5
 - (vi) their sum is 10
 - (vii) their product is 10
 - (viii) both are perfect numbers
 - (ix) both the numbers are less than 11 and having a difference of 2
 - (x) sum of the squares of both the numbers is a square of another number.
10. Two balls are drawn at random from a bag containing 4 red and 6 black balls. Find the probability that
- (i) both are red
 - (ii) both are black
 - (iii) one is red and another one is black
 - (iv) either both are red or both are black.
11. A natural number x is selected at random from first 100 natural numbers. Find the probability that x satisfies the equation(s)
- (i) $x^2 - 3x \leq 30$
 - (ii) $\frac{(x-40)(x-60)}{(x-20)} < 0$
 - (iii) $x^2 - 16 \geq 0$ and $x^2 - 81 \leq 0$.
12. In a hand of bridge, what is the chance that the 4 kings are held by a specified player?
13. If 12 persons be seated at a round table. What is the chance that 2 particular persons sit together?
14. What is the chance of getting a total of less than 12 on throwing a die two times?
15. 6 boys and 6 girls sit in a row randomly. Find the probability that all 6 girls sit together.
16. There are 5 addressed envelopes corresponding to 5 letters. If the letters are placed in the envelope at random, what is the probability that all the letters are not placed in the right envelopes?
17. The letters of the word IIT are placed at random. Find the probability that 2 vowels come together.
18. In a random arrangement of the letters of the word AIEEE, what is the chance that 3 Es come together.
19. There are 6 persons are seated at a row. Find the probability that 2 persons always sit together.
20. In a science conference hall, there are 5 Indian Scientists, 10 American Scientists and 8 Latin American Scientists seated in a row. What is the chance that all persons of the same nationality will sit together?
21. In a class, there are 6 boys and 4 girls seated in a row. Find the probability that no boy is in between two girls.
22. In a different arrangement of the letters of the word BANANA, what is the chance that two Ns do not appear adjacently?
23. In a random arrangement of the letters of the word SUCCESS, find the probability that all Ss do not come together.
24. In a class, there are 4 girls and 6 boys. Find the probability that they will be seated in a row so that all the 4 girls are not together.
25. In a class, there 10 boys and 9 girls. Find the probability that they will be seated in a row so that they are alternate.
26. In a class, 5 boys and 5 girls. Find the probability that they will be seated in a row so that boys and girls are alternate.
27. A bag contains 50 tickets numbered $\{1, 2, 3, \dots, 50\}$ of which 5 are drawn at random and arranged in ascending order of magnitude ($x_1 < x_2 < x_3 < x_4 < x_5$). Find the probability that $x_3 = 30$.
28. Find the probability that in a random arrangement of the letters of the word UNIVERSITY, so that two Is come together.

29. Find the probability that the birthdays of six different persons will fall in exactly two calendar months.
30. What is the probability that in a group of n persons, at least two of them will have the same birthday?
31. The odds in favour of an event are 3 : 5. Find the probability of occurrence of this event.
32. A card is drawn from an ordinary pack of 52 cards and a gambler bets that, it is a spade or an ace. What are the odds against his winning this bet?
33. Find the probability that a leap year selected randomly have 53 sundays?
34. Find the probability that a non-leap year selected randomly will have 53 sundays?
35. Find the probability that in a year of 22nd Century chosen at random there will be 53 sundays?
36. A natural number x is chosen at random from first 100 natural numbers such that

$$\frac{(x-10)(x-20)}{(x-30)} < 0$$

Find the probability of x .

37. A natural number x is chosen at random from $\{1, 2, \dots, 10\}$ such that $x^3 - 6x^2 + 11x - 6 = 0$. Find the probability of x .
38. Two distinct natural numbers are chosen at random from the first 100 natural numbers. Find the probability that
 - (i) the first one is divisible by 2 and the second one is divisible by 3
 - (ii) both are divisible by 2 or 3
 - (iii) both are divisible by 2 and 3
 - (iv) both are neither divisible by 2 nor 3
 - (v) the two numbers x and y satisfy the equation $3x - 4y = 0$.
39. If two of the 64 squares are chosen at random on a chess board. Find the probability that they have one side in common.
40. In a single throw of three dice, find the probability of getting a total of at least 5.
41. In a single throw of three dice, find the probability of getting a total of at least 15.
42. A determinant is chosen at random from the set of all determinants of order 2 with elements 0 or 1 only. Find the probability that the determinant chosen has the value non-negative.
43. Three dice are thrown. Find the probability of getting a sum, which is a perfect square.
44. Three distinct numbers are selected at random from the set $A = \{1, 2, \dots, 10\}$. Find the probability that the product of two of the numbers is equal to the third.
45. The interior angles of a regular polygon is 150° each. Find the probability of getting of the number of diagonals of the given regular polygon.
46. A function is selected at random from all the functions defined over a given set A consisting of 3 distinct elements. Find the probability that the mapping selected is one-one.
47. In a convex hexagon, two diagonals are drawn at random. Find the probability that the diagonals intersect at an interior point of the hexagon.
48. Three of the six vertices of a regular hexagon are chosen at random. Find the probability that the triangle formed by these vertices is an equilateral.
49. A mapping is selected at random from the set of all mappings of the set $A = \{1, 2, \dots, n\}$. Find the probability that the mapping selected is bijective.
50. A mapping is selected at random from $f: \{x_1, x_2, x_3, x_4, x_5\} \rightarrow \{y_1, y_2, y_3, y_4, y_5\}$ such that $f(x_i) \neq y_i, i = 1, 2, 3, 4, 5$. Find the probability that the selected mapping is one-one.
51. Let a set A has 7 elements and the set B has 5 elements. If one function is selected from all possible defined functions from A to B , find the probability that it is onto.
52. Of all the mappings that can be defined from the set $A = \{1, 2, 3, 4\}$ to the set $B = \{5, 6, 7, 8, 9\}$, a mapping is selected at random. Find the probability that the selected mapping is strictly monotonic
53. A die is rolled three times. Find the probability of getting a larger number than the previous number.
54. Four small squares on a chessboard are selected at random. Find the probability that they form a square of the size 2×2 .
55. Three letters are to be sent to different persons and addresses on the three envelopes are also be written without looking at the address in the letters. Find the probability that all the letters are not placed into the right envelop.
56. If three distinct numbers are chosen randomly from the first 100 natural numbers. Find the probability that all the three of them are divisible by both 2 and 3.

ADDITION THEOREM ON PROBABILITY

57. A natural number is chosen at random from first 100 natural numbers. Find the probability that it is divisible by 2 or 3.
58. A natural number is chosen at random from first 200 natural numbers. Find the probability that it is divisible by 3 or 4 or 5.
59. A card is drawn at random from a pack of 52 cards. Find the probability that it is a court card or a face card.
60. A card is drawn from a pack of 52 cards. Find the probability that the card drawn is a king or a heart card or a red card.
61. A die is thrown twice. Find the probability that the sum of the numbers of the two faces is divisible by 4 or 5.
62. A die is thrown two times. Find the probability that at least one of them shows the number 3.
63. A die is thrown two times. Find the probability of getting an odd number on the first die or a total of 7.

64. If two dice are thrown simultaneously, find the probability of the sum of the numbers coming up is greater than 9.
65. The probability of a student selected in IIT is $1/3$ and in AIEEE is $1/2$. Find the probability that he/she will get selected either of them.
66. Four cards are drawn from a pack of 52 cards. Find the probability that the drawn cards are of the same suit.
67. The probability of a student passing in mathematics is $1/3$ and in physics is $1/4$ and the probability of passing in both physics and mathematics is $1/5$. Find the probability that the student will pass at least one of the subjects and none of the subject.
68. Two cards are drawn from a pack of 52 cards. What is the probability that either both are red or both are kings?
69. The probability that a person will get an electric contract is $2/5$ and the probability that he will not get plumbing contract is $4/7$. If the probability of getting at least one contract is $2/3$, what is the probability that he will get both the contracts?
70. The probability that Hameed passes in mathematics is $2/3$ and the probability that he passes in English is $4/9$. If the probability of passing in both the courses is $1/4$, what is the probability that Hameed will pass in at least one of these subjects?
71. The probability that a person will travel by plane is $3/5$ and that he will travel by train is $1/4$. What is the probability that he (she) will travel by plane or train?
72. A card is drawn from a deck of 52 cards. Find the probability of getting a king or a heart or a red card.
73. Two cards are drawn at random from a deck of 52 cards. Find the probability of getting both the cards spade, heart or club.
74. An integer is chosen at random from the numbers ranging from 1 to 50. What is the probability that the integer chosen is a multiple of 2 or 3 or 10?
75. What is the chance of getting 4 or 7 or 12 in a throw of two dice?
76. Let A, B, C be three events. If the probability of occurring exactly one event out of A and B is $1 - x$, out of B and C is $1 - 2x$, out of C and A is $1 - x$, and that of occurring three events simultaneously is x^2 , prove the probability that at least one out of A, B, C will occur is greater than $1/2$.

INEQUALITY

77. If $P(A) = \frac{3}{5}$ and $P(B) = \frac{2}{3}$,

prove that

- (i) $P(A \cup B) \geq \frac{2}{3}$
- (ii) $\frac{4}{15} \leq P(A \cap B) \leq \frac{3}{5}$.

78. If $P(A) = \frac{3}{4}$ and $P(B) = \frac{3}{8}$,

prove that

- (i) $P(A \cup B) \geq \frac{1}{8}$
- (ii) $\frac{1}{8} \leq P(A \cap B) \leq \frac{3}{4}$.

CONDITIONAL PROBABILITY

79. A die is rolled. If the outcome is an odd number, find the probability that it is a prime.
80. A pair of dice is thrown. Find the probability of getting 7 as the sum, if it is known that the second die always exhibits a prime number.
81. Two dice are thrown. Find the probability that sum of the numbers coming up on them is 9, if it is known that the number 5 always occurs on the first die.
82. Two coins are tossed. What is the chance of coming up of two heads, if it is known that at least one head comes up?
83. A pair of dice is thrown. Find the probability of getting the sum 8 or more, if 4 appears on the first die.
84. A pair of dice is thrown. Find the probability of getting 7 as the sum, if it is known that the second die always exhibits a prime number.
85. A pair of dice is thrown. Find the probability of getting 7 as the sum, if it is known that the second die always exhibits an odd number.
86. Two dice are thrown. Find the probability that the numbers appeared has the sum 8, if it is known that the second die always exhibits 4.
87. A dice is thrown twice and the sum of the numbers appearing is observed to be 6. What is the conditional probability that the number 4 has appeared at least once?
88. To test the quality of electric bulbs produced in a factory, two bulbs are randomly selected from a large sample without replacement. If either bulb is defective, the entire lot is rejected. Suppose a sample of 200 bulbs contains 5 defective bulbs. Find the probability that the sample will be rejected.
89. A bag contains 10 white and 15 black balls. Two balls are drawn in succession without replacement. What is the probability that the first ball is white and the second is black?
90. Find the probability of drawing a diamond card in each of the two consecutive draws from a well-shuffled pack of cards, if the card drawn is not replaced after the first draw.
91. A bag contains 5 white, 7 red and 8 black balls. If four balls are drawn one by one without replacement, find the probability of getting all white balls.
92. Two integers are selected at random from integers 1 through 11. If the sum is even, find the probability that both the numbers are odd.

93. A couple has 2 children. Find the probability that both are boys, if it is known that
 (i) one of the children is a boy
 (ii) the older child is a boy.
94. Two coins are tossed. What is the probability of coming up two heads if it is known that at least one head comes up?
95. A card is drawn at random from a pack of 52 cards. What is the chance that it is a king, if it is known that the red card is drawn.

INDEPENDENT EVENTS

96. A bag contains 3 white and 2 black balls and another bag contains 3 white and 5 black balls. If one ball is drawn from each bag, find the probability that
 (i) both are white
 (ii) both are black
 (iii) one is white and one is black.
97. A problem in mathematics is given to 3 students whose chances of solving it are $\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{4}$ respectively. What is the probability that the problem is solved?
98. A speak truth in 75% cases and B in 80% of the cases. In what percentage of the cases are they likely to contradict each other in stating the same fact?
99. Three students appear at an examination of JEE-Advanced. The probabilities of their success are $\frac{1}{3}$, $\frac{1}{4}$, $\frac{1}{5}$ respectively. Find the probability of success of at least two.
100. A person is known to hit a target 3 out of 4 times and another person hits the target 2 out of 3 times. Find the probability that the target will be hit when they both try.
101. A speaks truth in 60% of the cases and B in 90% of the cases. In what percentage of cases are they likely to contradict each other in stating the same fact?
102. A can solve 90% of the problems given in a book and B can solve 70%. What is the probability that at least one of them will solve the problem selected at random from the book?
103. The probabilities of solving a specific problem independently by A and B are $\frac{1}{2}$ and $\frac{1}{3}$ respectively. If both try to solve the problem independently, find the probability that
 (i) the problem is solved
 (ii) exactly one of them solves the problem.
104. Given the probability that A can solve a problem is $\frac{2}{3}$ and the probability that B can solve the same problem is $\frac{3}{5}$. Find the probability that none of the two will be able to solve the problem.
105. Three cards are drawn with replacement from a well-shuffled pack of cards. Find the probability that the cards drawn are king, queen and jack.
106. The probability that A hits a target is $\frac{1}{3}$ and the probability that B hits it, is $\frac{2}{5}$. What is the probability that the target will be hit, if each one of A and B shoots at the target?

107. A bag contains 5 white, 7 red and 4 black balls. If four balls are drawn one by one with replacement, what is the probability that none is white?
108. A bag contains 5 white, 7 red and 8 black balls. Four balls are drawn one by one with replacement, what is the probability that at least one is white?
109. A can hit a target 4 times in 5 shots, B can hit 3 times in 4 shots, and C can hit 2 times in 3 shots. Calculate the probability that
 (i) A , B , C all may hit
 (ii) B , C may hit and A may lose.
110. A box contains 3 red and 5 blue balls. Two balls are drawn one by one at a time at random without replacement. Find the probability of getting 1 red and 1 blue ball.
111. Two cards are drawn from a well-shuffled pack of 52 cards without replacement, What is the probability that one is a red queen and the other is a black king?
112. Two cards are drawn without replacement from a well-shuffled pack of 52 cards. Find the probability that one is a spade and other is a queen of red colour.
113. Cards are numbered 1 to 25. Two cards are drawn one after the other. Find the probability that the number on one card is a multiple of 7, on the other it is a multiple of 11.
114. The probability of A , B , C solving a problem are $\frac{1}{3}$, $\frac{2}{7}$ and $\frac{3}{8}$, respectively. If all the three try to solve the problem simultaneously, find the probability that exactly one of them can solve it.
115. The probability of a student A passing an examination is $\frac{3}{7}$ and of the student B passing is $\frac{5}{7}$. Assuming the two events ' A passes', ' B passes', as independent, find the probability
 (i) only A passing the examination
 (ii) only one of them passing the examination.
116. Two persons A and B throw a die alternately till one of them gets a 'three' and wins the game. Find their respective probabilities of winning, if A begins.
117. A and B throw alternately with a pair of dice. A wins if he throws 6 before B throws 7 and B wins if he throws 7 before A throws 6. Find the chance of winning if A makes the first throw.
118. Three persons A , B , C throw a die in succession till one gets a 'six' and wins the game. Find their respective probabilities of winning, if A begins.

TOTAL PROBABILITY

119. A bag contains 4 red and 3 black balls. A second bag contains 2 red and 4 black balls. One bag is selected at random. From the selected bag, one ball is drawn. Find the probability that the ball drawn is red.
120. One bag contains 4 white and 5 black balls. Another bag contains 6 white and 7 black balls. A ball is trans-

- ferred from the first bag to the second bag and then a ball is drawn from the second bag. Find the probability that the ball drawn is white.
121. A lot contains 20 articles. The probability that the lot contains exactly two defective articles is 0.4 and the probability that the lot contains exactly 3 defective articles is 0.6. Articles are drawn from the lot at random one by one, without replacement and are tested till all the defective articles are found. What is the probability that the testing procedure ends at the 12th testing?
 122. In a bolt factory, machines A , B and C manufacture respectively 25%, 35% and 40% of the total bolts. Of their output 5, 4 and 2 per cent are, respectively, defective bolts. A bolt is drawn at random from the product, what is the probability that the bolt drawn is defective?
 123. Two-thirds of the students in a class are boys and the rest are girls. It is known that the probability of a girl getting a first class is 0.25 and that of a boy getting a first class is 0.28. Find the probability that a student chosen at random will get first class.
 124. An urn contains 2 white and 2 black balls. A ball is drawn at random. If it is white, it is not replaced into the urn. Otherwise it is replaced along with another ball of the same colour. The process is repeated. Find the probability that the third ball drawn is black.
 125. An unbiased coin is tossed. If the result is head, a pair of unbiased dice is rolled and the number obtained by adding the number on the two faces is noted. What is the probability that the noted number is either 7 or 8?
 126. An urn contains m white and n black balls. A ball is drawn at random and is put into the urn along with k additional balls of the same colour as that of the ball drawn. A ball is again drawn at random. What is the probability that the ball drawn now is white?
- BAYE'S THEOREM**
127. A bag contains 3 red and 4 black balls and 2nd bag contains 5 red and 6 black balls. One ball is drawn at random from one of the bags and is found to be red. Find the probability that it was drawn from second bag.
 128. An insurance company insured 2000 scooter drivers, 4000 car drivers and 6000 truck drivers. The probability of an accident involving a scooter driver, car driver and truck driver is 0.01, 0.03 and 0.15, respectively. One of the insured person meets an accident. What is the probability that he is a scooter driver?
 129. A man is known to speak truth 3 times out of 5 times. He throws a die and reports that it is a six. Find the probability that it is actually six.
 130. A speaks the truth 3 out of 4 times, and B speaks truths 5 out of 6 times. What is the probability that they will contradict each other in stating the same fact?
 131. A speaks the truth 2 out of 3 times and B speaks truths 4 times out of 5. They agree in the assertion that from a bag containing 6 balls of different colors a red ball has been drawn. Find the probability that the statement is true.
 132. A company has two plants to manufacture bicycles. The first plant manufactures 60% of the bicycles and the second plant 40%. Out of that 80% of the bicycles are rated of standard quality at the first plant and 90% of standard quality at the second plant. A bicycle is picked up at random and found to be of standard quality. Find the probability that it comes from the second plant.
 133. In a test, an examinee either guess or copies or knows the answer to a multiple-choice questions with four choices. The probability that he makes a guess is $\frac{1}{3}$ and the probability that he copies the answer is $\frac{1}{6}$. The probability that his answer is correct, given that he copied it is $\frac{1}{8}$. Find the probability that he knew the answer to the question, given that he correctly answered it.
 134. A pack of playing cards was found to contain only 51 cards. If the first 13 cards which are examined are all red. What is the probability that the missing card is a black card.
 135. An employer sends a letter to his employee but he does not receive the reply. (It is certain that the employee would had replied it if he did receive the letter.) It is known that one out of n letters does not reach the destination. Find the probability that the employee does not receive the letter.
 136. In a combat between A , B and C , A tries to hit B and C and B and C try to hit A . The probabilities of A , B and C hitting the targets are $\frac{2}{3}$, $\frac{1}{2}$ and $\frac{1}{3}$ respectively. If A is hit, find the probability that B hits and A and C does not.
 137. A box contains N coins, m of which are fair and the rest are biased. The probability of getting a head when a fair coin is tossed is $\frac{1}{2}$, while it is $\frac{2}{3}$ when a biased coin is tossed. A coin is drawn from the box at random and is tossed twice. The first time it shows a head and the second time it shows a tail. What is the probability that the coin drawn is fair?
 138. A bag contains 6 white and 12 red balls. Six balls are drawn at random without replacement. If at least four of these are white, find the probability that in the next two draws exactly one white ball is drawn.
 139. A person goes to the office either by a car or by a scooter or by bus or by train. The probabilities of his using car, scooter, bus and train are respectively, $\frac{1}{2}$, $\frac{2}{7}$, $\frac{3}{7}$ and $\frac{1}{7}$. The probabilities of his reaching late in the office by using these modes of transport are $\frac{2}{9}$, $\frac{4}{9}$, $\frac{1}{9}$ and $\frac{1}{9}$, respectively. If the person reaches the office in time, find the probability that he used the car to reach the office.
 140. A card from a pack of 52 cards is lost. From the remaining cards of the pack two cards are drawn at random and are found to be spade. Find the probability that the missing card to be a spade.

141. A speaks truth 3 times out of 5 while B speaks truths 7 times out of 10. A ball is drawn at random from a bag containing one black ball and 5 other balls of different colors. Both A and B report that a black ball has been drawn from the bag. Find the probability of their assertion being true.
142. A letter is known to have come from CALCUTTA or TATANAGAR. On the envelope, just two consecutive letters TA are visible. Find the probability that the letter came from CALCUTTA.
143. A letter is known to have come from either MAHARASTRA or MADRAS. On the post mark only consecutive letters RA can be read clearly. What is the chance that the letter came from MAHARASTRA?
144. A can hit a target 4 times in 5 shots, B can hit 3 times in 4 shots and C twice in 3 shots. They fire once each. If two of them hit. What is the chance that C has missed it?
145. A and B are two independent witnesses (i.e. there is no collusion between them). The probability that A and B will speak truths are x and y , respectively. A and B agree in a certain statement. Prove that the probability that the statement is true is $\frac{xy}{1 - x - y + xy}$.
146. There are two bags, one of which contains 3 black and 4 white balls and the other contains 4 black and 3 white balls. A ball is taken out from the first bag, and a ball is taken from the second bag. Find the probability of getting a black ball and that was drawn from the second bag.

PROBABILITY DISTRIBUTION

147. Find the probability distribution of X , the number of heads in two tosses of a coin (or a simultaneous toss of two coins).
148. An urn contains 4 white and 6 red balls. 4 balls are drawn at random from the urn. Find the probability distribution of the number of white balls.
149. An unbiased die is thrown twice. Find the probability distribution of the number of sixes.
150. Two cards are drawn successively with replacement from a well-shuffled pack of 52 cards. Find the probability distribution of the number of kings.
151. Three cards are drawn from a pack of 52 playing cards. Find the probability distribution of the number of aces.
152. From a lot of 30 bulbs which includes 6 defectives, a sample of 4 bulbs is drawn at random with replacement. Find the probability distribution of the number of defective bulbs.
153. A random variable X takes the values $-3, -2, 0, 1, 2$ and 3 such that

$$\begin{aligned} P(X=0) &= P(X>0) = P(X<0) \\ P(X=-3) &= P(X=-2) = P(X=-1) \\ P(X=1) &= P(X=2) = P(X=3) \end{aligned}$$

Obtain the probability distribution of X .

154. Let X be a random variable which assumes values x_1, x_2, x_3, x_4 such that
 $2P(X=x_1) = 3P(X=x_2) = P(X=x_3) = 5P(X=x_4)$.
 Find the probability distribution of X .

BINOMIAL DISTRIBUTION

155. A die is thrown 6 times. If getting a number 6 is a success, find the probability that
 (i) at least 5 successes
 (ii) at most 5 successes
 (iii) exactly 5 successes
 (iv) no success.
156. A die is thrown 6 times. If 'getting an odd number' is a 'success', what is the probability of
 (i) 5 successes
 (ii) at least 5 successes
 (iii) at most 5 successes
 (iv) at least one success
 (v) no success.
157. A pair of dice is thrown 7 times. If getting a total of 7 is a success, what is the probability of
 (i) no success
 (ii) 6 successes
 (iii) at least 6 successes
 (iv) at most 6 successes
158. Two fair dice are thrown simultaneously. If the operation is repeated 5 times, find the probability that the sum will be 7 exactly 4 times.
159. The probability that Pakistan wins a cricket match against India is $1/5$. If India and Pakistan play 5 test matches, what is the probability that India will win 4 test matches?
160. A pair of fair dice is thrown 4 times. Find the probability of getting a doublet at least 2 times.
161. If a coin is tossed n times. Find the probability that the head will appear an odd number of times.
162. The probability that a student is not a cricketer is $1/10$. Find the probability that out of 10 students, 7 are cricketer.
163. The probability that a student can get a chance in IIT is $3/4$. He tries 5 times to get a chance. Find the probability that he tries to get a chance in IIT at least 3 times.
164. The probability of guessing correctly straight objective-type multiple choice questions in IIT is $2/7$. Find the probability that a student can give right answer at least 8 questions out of 10 questions.
165. A family has 5 children. Assuming that the probability of a male child and a female child is equal. Find the probability that the family has at least 2 male children.
166. If n chocolates are distributed among m students in MIT-JEE Institute. Find the probability that a particular student will get r ($< n$) chocolates.
167. A man takes a step forward with probability 0.4 and backward with probability 0.6. Find the probability that at the end of eleven steps he is just one step away from starting point.

168. The probability of a man hitting a target is $1/4$. How many times must he fire so that the probability of his hitting the target at least once is greater than $2/3$?
169. How many dice must be thrown so that there is a better chance of obtaining a six?
170. If m things are distributed among a men and b women, show that the probability that the number of things received by men is odd is $\frac{1}{2} \times \left\{ \frac{(b+a)^m - (b-a)^m}{(b+a)^m} \right\}$.
171. Suppose the probability for A to win a game against B is 0.4 . If A has an option of playing a 'best of 3 games' or a 'best of 5 games' match against B , which option should A choose so that the probability of his winning the match is higher? (No game ends in a draw)
172. Find the maximum number of tosses of a pair of dice so that the probability of getting the sum of the digits on the dice equal to 7 on at least one toss is greater than 0.95 .
173. A die is thrown $(2n+1)$ times. Prove that the probability of getting 1, 3 or 5 at most n times is $1/2$.
174. In a hurdle race, a player has to cross 10 hurdles. The probability that he will clear each hurdle is $5/6$. Find the probability that he will knock down fewer hurdles.
175. Find the mean, variance and the standard deviation of the binomial distribution with parameters 4 and $1/3$.
176. The mean and the variance of a binomial distribution are 4 and $4/3$ respectively, find $P(X \geq 1)$.
177. If the sum of the mean and the variance of a binomial distribution for 5 trials is 1.8, find the distribution.
- GEOMETRICAL PROBABILITY**
178. A telephone subscriber has ordered a call. He may get it at any time in the next hour. What is the probability of the call taking place in the last 20 minutes of the hour?
179. Two points are selected in a line AB of length a so as to lie in opposite sides of its middle point C . Find the probability that the distance between them is less than $a/3$.
180. Two points are chosen independently at random from the interval $[0, 1]$. What is the chance that the two numbers differ by more than $1/2$?
181. Two numbers are selected independently at random from the interval $[0, 1]$. If the smaller one is less than $1/3$, find the probability that the larger one is greater than $3/4$ is $3/10$.
182. A line is divided at random into three parts. What is the chance that they form the sides of a possible triangle?
183. Let the sum of two positive numbers be 24. Find the probability that the product of two numbers is not less than $3/4$ times of their greatest product.
184. A circular target is divided into three zones bounded by concentric circles of radii $1/3$, $1/2$ and 1 respectively. If three shots are fixed at random at the target, what is the probability that exactly one shot lands in each zone?
185. If p is chosen at random in $[0, 5]$, what is the probability of the equation $x^2 + px + \frac{p+2}{4} = 0$ to have real roots?
186. Two real numbers x and y are selected at random, given that $0 \leq x \leq 1$, $0 \leq y \leq 1$. Find the probability $y^2 \leq x$.
187. Two persons A and B agree to meet at a place between 11 to 12 noon. The first one to arrive, wait for 20 minutes and then leave. If the time of their arrival be independent and at random, what is the probability that A and B meet?
188. Two real numbers x and y are chosen at random and are such that $|x| \leq 3$, $|y| \leq 5$. What is the probability that the fraction x/y being positive?
189. A real number x is selected at random from the solution set $y^2 - y - 6 \leq 0$. What is the probability that $\frac{(x+1)(x-2)}{(x-4)} \geq 0$?
190. Three points P, Q, R are selected at random from the circumference of a circle. Find the probability that the points lie on a semicircle.
191. Two real numbers x and y are randomly chosen on the real number line, such that $0 \leq x, y \leq 0$. Find the probability that the difference between the chosen numbers is not greater than 10.
192. On a straight line of length a , two points are taken at random. Find the probability that the distance between them is greater than b .
193. Two positive numbers x and y are chosen at random for which, each of the two does not exceed 2. Find the probability that $xy \leq 1, \frac{y}{x} \leq 2$.
194. Two friends decide to meet at a spot between 2 p.m. and 3 p.m. Whosoever arrives first agrees to wait for 15 minutes for the other. What is the probability that they meet?

Hints & Solutions

Level - 1

1. Here, $S = \{HH, HT, TH, TT\}$.
Then,

(i) the probability of 2 heads = $\frac{1}{4}$

(ii) the probability of exactly one head, i.e.

$$\{HT, TH\} \text{ is } = \frac{2}{4} = \frac{1}{2}.$$

(iii) the probability of exactly two tails, i.e.

$$\{TT\} = \frac{1}{4}.$$

2. Here,

$$S = \{HHH, HHT, HTH, THH, TTT, TTH, THT, HTT\}$$

Then

(i) the probability of exactly two heads, i.e.

$$\{HHT, HTH, THH\} = \frac{3}{8}$$

(ii) the probability of at least two heads, i.e.

$$\{HHT, HTH, THH, HHH\} = \frac{4}{8} = \frac{1}{2}$$

(iii) the probability of exactly one head, i.e.

$$\{TTH, THT, HTT\} = \frac{3}{8}.$$

3. Here, $S = \{(1, 1), (1, 2), (1, 3), \dots, (6, 6)\}$

(i) Let A be the event, which shows the outcome, a sum of 10 = $\{(4, 6), (6, 4), (5, 5)\}$

So, the probability of A,

$$P(A) = \frac{3}{36} = \frac{1}{12}$$

(ii) Let B be the event, which shows the outcome, a sum of at least 9.

= {a sum of 9, 10, 11 and 12}

= $\{(3, 6), (6, 3), (4, 5), (5, 4), (4, 6), (6, 4), (5, 5), (5, 6), (6, 5), (6, 6)\}$

So, the probability of B,

$$P(B) = \frac{10}{36} = \frac{5}{18}$$

(iii) Let C be the event which shows the outcome, a sum of even numbers

= {a sum of 2, 4, 6, 8, 10, 12}

= $\{(1, 1), (1, 3), (3, 1), (2, 2), (1, 5), (5, 1), (2, 4), (4, 2), (3, 3), (2, 6), (6, 2), (3, 5), (5, 3), (4, 4)\}$

So, the probability of C,

$$P(C) = \frac{18}{36} = \frac{1}{2}$$

(iv) Let D be the event shows the outcome, a sum of odd numbers

= {a sum of 3, 5, 7, 9, 11}

= $\{(1, 2), (2, 1), (1, 4), (4, 1), (2, 3), (3, 2), (1, 6), (6, 1), (2, 5), (5, 2), (3, 4), (4, 3), (3, 6), (6, 3), (4, 5), (5, 4), (5, 6), (6, 5)\}$

So, the probability of D,

$$P(D) = \frac{18}{36} = \frac{1}{2}$$

(v) Let E be the event. A sum of perfect numbers = {a sum of 6} (since only two perfect number exist from 1 to 100 natural numbers, say, 6 and 28)

= $\{(1, 5), (5, 1), (2, 4), (4, 2), (3, 3)\}$

So, the probability of E ,

$$P(E) = \frac{5}{36}$$

- (vi) Let F be the event, both the dice show prime numbers = $\{2, 3, 5\}$

$$= \{(2, 3), (2, 5), (5, 2), (3, 2), (3, 5), (5, 3)\}$$

So, the probability of F ,

$$P(F) = \frac{6}{36} = \frac{1}{6}$$

4. Here,

$$S = \{(1, 1, 1), (1, 1, 2), (1, 1, 3), \dots, (6, 6, 6)\}$$

- (i) Let A be the event of getting a sum of 3, i.e.

$$\{(1, 1, 1)\}$$

So, the probability of A ,

$$P(A) = \frac{1}{216}$$

- (ii) Let B be the event of getting a sum of 4 $\rightarrow (1, 1, 2)$

These three numbers can be arranged themselves

in $\frac{(3)!}{(2)!}$ ways, i.e. in 3 ways

Hence, the probability of B ,

$$P(B) = \frac{3}{216} = \frac{1}{72}$$

- (iii) Let C be the event of getting a sum of 5 $\rightarrow (1, 1, 3), (1, 2, 2)$

Hence, the probability of C ,

$$P(C) = \frac{3+3}{216} = \frac{6}{216} = \frac{1}{36}$$

- (iv) Let D be the event of getting a sum of 6 $\rightarrow (1, 2, 3), (1, 1, 4), (2, 2, 2)$

Hence, the probability of D ,

$$P(D) = \frac{3+6+3}{216} = \frac{12}{216} = \frac{1}{18}$$

- (v) Let E be the event of getting a sum of 7 $\rightarrow (1, 1, 5), (1, 2, 4), (1, 3, 3), (2, 2, 3)$

Hence, the probability of E ,

$$P(E) = \frac{3+6+3+3}{216} = \frac{15}{216} = \frac{5}{72}$$

- (vi) Let F be the event of getting a sum of at least 15 = a sum of $\{15, 16, 17, 18\}$

$$15 \rightarrow (3, 6, 6), (4, 5, 6), (5, 5, 5),$$

$$16 \rightarrow (4, 6, 6), (5, 5, 6),$$

$$17 \rightarrow (5, 6, 6)$$

$$\text{and } 18 \rightarrow (6, 6, 6)$$

Hence, the required probability,

$$P(F) = \frac{10+6+3+1}{216} = \frac{20}{216} = \frac{5}{54}$$

- (vii) Let G be the event of getting

A sum of at least

$$16 = \{16, 17, 18\}$$

$$16 \rightarrow (4, 6, 6), (5, 5, 6),$$

$$17 \rightarrow (5, 6, 6)$$

$$\text{and } 18 \rightarrow (6, 6, 6)$$

Hence, the required probability,

$$P(G) = \frac{6+3+1}{216} = \frac{10}{216} = \frac{5}{108}$$

5. (i) Required probability of a red card = $\frac{26}{52} = \frac{1}{2}$

- (ii) Required probability of a black card = $\frac{26}{52} = \frac{1}{2}$

- (iii) Required probability of a king card = $\frac{4}{52} = \frac{1}{13}$

- (iv) Required probability of a king of a red colour = $\frac{2}{52} = \frac{1}{26}$

- (v) Required probability of a queen of spade = $\frac{1}{52}$

- (vi) Required probability of a face card = $\frac{12}{52} = \frac{3}{13}$

6. (i) Required probability of getting both are red cards

$$= \frac{{}^{26}C_2}{{}^{52}C_2} = \frac{26 \times 25}{52 \times 51} = \frac{25}{102}$$

- (ii) Required probability of getting both are king cards

$$= \frac{{}^4C_2}{{}^{52}C_2} = \frac{4 \times 3}{52 \times 51} = \frac{1}{13 \times 17} = \frac{1}{221}$$

- (iii) Required probability of getting one is a face card and another one is a king of a red colour.

$$= \frac{{}^{12}C_1 \times {}^2C_1}{{}^{52}C_2} = \frac{12 \times 2 \times 2}{52 \times 51} = \frac{12}{13 \times 51} = \frac{12}{663}$$

- (iv) Required probability of getting one is a court card and another one is an ace card

$$= \frac{{}^{16}C_1 \times {}^4C_1}{{}^{52}C_2} = \frac{16 \times 4 \times 2}{52 \times 51} = \frac{32}{13 \times 51} = \frac{32}{663}$$

- (v) Required probability of getting both are of the same suit.

$$= \frac{{}^{13}C_2 + {}^{13}C_2 + {}^{13}C_2 + {}^{13}C_2}{{}^{52}C_2}$$

$$= \frac{4 \times {}^{13}C_2}{{}^{52}C_2} = \frac{4 \times 13 \times 12}{52 \times 51} = \frac{4}{13}$$

- (vi) Required probability of getting both are of the different suit

$$= \frac{4({}^{13}C_1 \times {}^{13}C_1 + {}^{13}C_1 \times {}^{13}C_1)}{{}^{52}C_2}$$

$$= \frac{169}{52 \times 51} = \frac{169 \times 2}{52 \times 51} = \frac{13}{102}$$

7. (i) Required probability of getting 2 red cards and 2 kings of black colour.

$$= \frac{{}^{26}C_2 \times {}^2C_2}{{}^{52}C_4}$$

- (ii) Required probability of getting all have come from the different suit

$$= \frac{{}^{13}C_1 \times {}^{13}C_1 \times {}^{13}C_1 \times {}^{13}C_1}{{}^{52}C_4}$$

- (iii) Required probability of getting the cards, all coming from the same suit

$$= \frac{{}^{13}C_4 + {}^{13}C_4 + {}^{13}C_4 + {}^{13}C_4}{{}^{52}C_4} = \frac{4 \times {}^{13}C_4}{{}^{52}C_4}$$

- (iv) Required probability of getting all the cards are honours cards

$$= \frac{{}^{12}C_4}{{}^{52}C_4}$$

- (v) Required probability of getting all are face cards

$$= \frac{{}^{12}C_4}{{}^{52}C_4}$$

- (vi) Required probability of getting 3 are court cards

$$= \frac{{}^{16}C_3 \times {}^{36}C_1}{{}^{52}C_4}$$

- (vii) Required probability of getting 2 are aces

$$= \frac{{}^4C_2 \times {}^{48}C_2}{{}^{52}C_4}$$

- (viii) Required probability of getting 2 slave cards

$$= \frac{{}^{12}C_2 \times {}^{40}C_2}{{}^{52}C_4}$$

- (ix) Required probability of getting all are number

$$\text{cards} = \frac{{}^{36}C_4}{{}^{52}C_4}$$

- (x) Required probability of getting all are same num-

$$\text{ber cards} = \frac{{}^4C_4 \times 9}{{}^{52}C_4}$$

8. (i) Required probability of getting an even number

$$= \frac{50}{100} = \frac{1}{2}$$

- (ii) Required probability of getting an odd number

$$= \frac{50}{100} = \frac{1}{2}$$

- (iii) Required probability of getting a prime number

$$= \frac{24}{100} = \frac{6}{25}$$

- (iv) Required probability of getting a perfect number

$$= \frac{2}{100} = \frac{1}{50}$$

- (v) Required probability of getting a number which

$$\text{have three factors only} = \frac{4}{100} = \frac{1}{25}$$

- (vi) Required probability of getting a number which have only 4-factors

$$= \frac{30}{100} = \frac{3}{10}$$

(those numbers are 6, 8, 10, 14, 15, 21, 22, 26, 27, 33, 34, 35, 38, 46, 51, 55, 57, 58, 65, 69, 74, 78, 82, 85, 86, 87, 89, 91, 94, 95)

- (vii) Required probability of getting a perfect square

$$= \frac{10}{100} = \frac{1}{10}$$

- (viii) Required probability of getting a perfect cube

$$= \frac{3}{100}$$

- (ix) Required probability of getting a perfect fourth

$$\text{power} = \frac{2}{100} = \frac{1}{50}$$

- (x) Required probability of getting a perfect sixth

$$\text{power} = \frac{2}{100} = \frac{1}{50}$$

9. Here, $S = \{1, 2, 3, 4, \dots, 100\}$

- (i) Let A be the event which is divisible by both 2 and 3

$$= \{6, 12, 18, \dots, 96\}$$

Then the probability of A ,

$$\begin{aligned} P(A) &= \frac{n(A)}{n(S)} = \frac{{}^{16}C_2}{{}^{100}C_2} \\ &= \frac{16 \times 15}{100 \times 99} = \frac{1}{5 \times 33} = \frac{1}{165} \end{aligned}$$

- (ii) Let B be the event which is divisible by both 3 and 4.

$$= \{12, 24, \dots, 96\}$$

Then the probability of B ,

$$\begin{aligned} P(B) &= \frac{n(A)}{n(S)} = \frac{{}^8C_2}{{}^{100}C_2} \\ &= \frac{8 \times 7}{100 \times 99} = \frac{7}{2475} \end{aligned}$$

- (iii) Let C be the event which is divisible by 3 and 7

$$= \{12, 24, 63, 84\}$$

Then the probability of C ,

$$\begin{aligned} P(C) &= \frac{n(A)}{n(S)} = \frac{{}^4C_2}{{}^{100}C_2} \\ &= \frac{4 \times 3}{100 \times 99} = \frac{1}{25 \times 33} = \frac{1}{825} \end{aligned}$$

- (iv) Let D be the event which is divisible by both 3 and 5.

$$= \{15, 30, \dots, 90\}$$

Then the probability of D ,

$$P(D) = \frac{n(A)}{n(S)} = \frac{{}^6C_2}{{}^{100}C_2} \\ = \frac{6 \times 5}{100 \times 99} = \frac{1}{330}$$

- (v) Let E be the event, whose first outcome is divisible by 2 and second outcome is divisible by 5.

Then the probability of E ,

$$P(E) = \frac{{}^{50}C_1 \times {}^{10}C_1}{{}^{100}C_2} = \frac{50 \times 10 \times 2}{100 \times 99} = \frac{10}{99}$$

- (vi) Let F be the event, whose sum is 10

$$= \{(1, 9), (2, 8), (3, 7), (4, 6), (5, 5), \\ (6, 4), (7, 3), (8, 2), (9, 1)\}$$

Then the probability of F ,

$$P(F) = \frac{{}^9C_2}{{}^{100}C_2} = \frac{9 \times 8}{100 \times 99} = \frac{2}{275}$$

- (vii) Let G be the event, whose product of the integer is 10

$$= \{(1, 10), (2, 5), (5, 2), (10, 1)\}.$$

Then the probability of G ,

$$P(G) = \frac{{}^4C_2}{{}^{100}C_2} = \frac{4 \times 3}{100 \times 99} = \frac{1}{825}$$

- (viii) Let H be the event, where both integers are perfect numbers

$$= \{6, 28\}$$

Then the probability of H ,

$$P(H) = \frac{{}^1C_1}{{}^{100}C_2} = \frac{1 \times 2}{100 \times 99} = \frac{1}{4950}$$

- (ix) Let I be the event, where both integers are less than 11 and their difference is 2

$$= \{(1, 3), (3, 1), (2, 4), (4, 2), (3, 5), \\ (5, 3), (4, 6), (6, 4), (5, 7), (7, 5), (7, 9), \\ (9, 7), (8, 10), (10, 8)\}$$

Then the probability of I ,

$$P(I) = \frac{{}^{12}C_2}{{}^{100}C_2} = \frac{12 \times 11}{100 \times 99} = \frac{3}{25 \times 9} = \frac{1}{75}$$

- (x) Let J be the event, where the sum of the squares of both the numbers is a square of another number

$$= \{(3, 4, 5), (6, 8, 10)\}$$

Then the probability of J ,

$$P(J) = \frac{2}{{}^{100}C_2} \\ = \frac{2 \times 2}{100 \times 99} = \frac{1}{25 \times 99} = \frac{1}{2475}$$

$$10. (i) \text{ Required probability} = \frac{{}^4C_2}{{}^{10}C_2} = \frac{4 \times 3}{10 \times 9} = \frac{2}{15}$$

$$(ii) \text{ Required probability} = \frac{{}^6C_2}{{}^{10}C_2} = \frac{6 \times 5}{10 \times 9} = \frac{1}{3}$$

$$(iii) \text{ Required probability} = \frac{{}^4C_1 \times {}^6C_1}{{}^{10}C_2} \\ = \frac{4 \times 6 \times 2}{10 \times 9} = \frac{2 \times 2 \times 2}{5 \times 3} = \frac{8}{15}$$

$$(iv) \text{ Required probability} = \frac{{}^4C_2 + {}^6C_2}{{}^{10}C_2} \\ = \frac{4 \times 3 + 6 \times 5}{10 \times 9} \\ = \frac{12 + 30}{90} = \frac{42}{90} = \frac{7}{15}$$

11. (i) Given in-equation is

$$x^2 - 3x \leq 10$$

$$\Rightarrow x^2 - 3x - 10 \leq 0$$

$$\Rightarrow (x - 5)(x + 2) \leq 0$$

$$\Rightarrow -2 \leq x \leq 5$$

$$\Rightarrow x = 1, 2, 3, 4, 5$$

Hence, the required probability

$$= \frac{5}{100} = \frac{1}{20}$$

$$(ii) \text{ Given in-equation is } \frac{(x-40)(x-60)}{(x-20)} < 0$$

$$\Rightarrow \frac{(x-40)(x-60)}{(x-20)} < 0$$

$$\Rightarrow x \in (1, 20) \cup (40, 60)$$

$$\text{Hence, the required probability} = \frac{37}{100}$$

- (iii) Given inequations are

$$x^2 - 16 \geq 0 \text{ and } x^2 - 81 \leq 0.$$

$$\Rightarrow (x+4)(x-4) \geq 0 \text{ and } (x+9)(x-9) \leq 0$$

$$\Rightarrow x \in [1, 4] \cup [1, 9]$$

$$\text{Hence, the required probability} = \frac{13}{100}.$$

$$12. \text{ Hence the required probability} = \frac{{}^4C_4 \times {}^{48}C_9}{{}^{52}C_{13}}$$

$$= \frac{{}^{48}C_9}{{}^{52}C_{13}}$$

$$13. \text{ Hence the required probability} = \frac{(11)! \times (2)!}{(12)!} = \frac{1}{6}$$

$$14 \text{ Hence, the required probability} = 1 - \frac{1}{36} = \frac{35}{36}$$

15. Hence, the required probability = $\frac{(7)! \times (6)!}{(12)!}$
16. Hence the required probability = $1 - \frac{1}{(5)!} = \frac{119}{120}$
17. Hence, the required probability = $\frac{(2)! \times (2)!}{(3)!} = \frac{4}{6} = \frac{2}{3}$
18. Hence, the required probability = $\frac{(3)! \times (3)!}{(5)!} = \frac{36}{120} = \frac{3}{10}$
19. Hence, the required probability is = $\frac{(5)! \times (2)!}{(6)!} = \frac{2}{6} = \frac{1}{3}$
20. Hence, the required probability = $\frac{(5)! \times (10)! \times (8)! \times (3)!}{(23)!}$
21. Hence the required probability = $\frac{(7)! \times (4)!}{(10)!}$
22. Hence, the required probability
- $$= \frac{(6)!}{(2)! \times (3)!} - \frac{(5)!}{(3)!} = \frac{60 - 20}{50} = \frac{40}{50} = \frac{4}{5}$$
23. Hence, the required probability
- $$= \frac{(7)!}{(2)! \times (3)!} \times \frac{(5)!}{(2)! \times (3)!} = \frac{7 \times 60 - 60}{7 \times 60} = \frac{6 \times 60}{7 \times 60} = \frac{6}{7}$$
24. Hence, the required probability = $\frac{(6)! \times {}^7P_4}{(10)!}$
25. Hence, the required probability = $\frac{(10)! \times (9)!}{(19)!}$
26. Hence, the required probability = $\frac{(5)! \times (5)! \times 2}{(10)!}$
27. Hence, the required probability = $\frac{{}^{29}C_2 \times {}^1C_1 \times {}^{20}C_2}{{}^{50}C_5}$
28. Hence, the required probability = $\frac{9! \times 2!}{10!} = \frac{2}{10} = \frac{1}{5}$
29. Six persons born in 12 months = 12^6 ways
Now, any two months can be chosen in ${}^{12}C_2$ ways

The six birthdays can fall in these two months in 2^6 ways. Out of these 2^6 ways, there are 2 ways when all the six birthdays fall in one month.

So, the number of possible cases = ${}^{12}C_2 \times (2^6 - 2)$

$$\begin{aligned} \text{Hence, the required probability} &= \frac{{}^{12}C_2 \times (2^6 - 2)}{12^6} \\ &= \frac{12 \times 11 \times (2^5 - 1)}{12^6} \\ &= \frac{11 \times 31}{12^5} = \frac{341}{12^5} \end{aligned}$$

30. There are 365 days in a year $n(S)$ = the number of ways in which n persons can have birthdays
 $= 365 \times 365 \times 365 \times \dots n$ times
 $= (365)^n$ ($\because$ each person can have birthday in 365 ways)
 $n(E)$ = the number of ways in which n persons can have different birthdays
 $= (365)^n - {}^{365}C_n$

$$\begin{aligned} \text{Hence, the required probability} &= \frac{(365)^n - {}^{365}C_n}{(365)^n} \\ &= \left(1 - \frac{{}^{365}C_n}{(365)^n} \right) \end{aligned}$$

31. Hence, the required probability = $\frac{3}{3+5} = \frac{3}{8}$

32. We know that, odds against of an event $E = \frac{n(E')}{n(E)}$

$$= \frac{P(E')}{P(E)} = \frac{9/13}{4/13} = \frac{9}{4},$$

where $P(E) = \frac{13}{52} + \frac{4}{52} - \frac{1}{52} = \frac{16}{52} = \frac{4}{13}$

and $P(E') = 1 - \frac{4}{13} = \frac{9}{13}$

33. Let $S = \{(\text{Mon, Tue}), (\text{Tue, Wed}), (\text{Wed, Thu}), (\text{Thu, Fri}), (\text{Fri, Sat}), (\text{Sat, Sun}), (\text{Sun, Mon})\}$
 and $A = \{(\text{Sat, Sun}), (\text{Sun, Mon})\}$

Thus, $P(A) = \frac{n(A)}{n(S)} = \frac{2}{7}$

34. Let $S = \{\text{Mon, Tue, Wed, Thu, Fri, Sat, Sun}\}$
 and $A = \{\text{Sun}\}$

Hence, the required probability = $P(A) = \frac{n(A)}{n(S)} = \frac{1}{7}$

35. Here, $P(\text{leap year}) = \frac{1}{4}$ and

$P(\text{non-leap year}) = \frac{3}{4}$

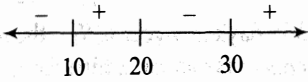
Let A and B be the events that a non-leap year and a leap year, respectively, have 53 Sundays.

Thus, $P(A) = \frac{1}{4} \times \frac{2}{7}$ and $P(B) = \frac{3}{4} \times \frac{1}{7}$

$$\begin{aligned}
 \text{Hence, the required probability} &= P(A \text{ or } B) \\
 &= P(A \cup B) \\
 &= P(A) + P(B) \\
 &= \frac{1}{4} \times \frac{2}{7} + \frac{3}{4} \times \frac{1}{7} \\
 &= \frac{5}{28}
 \end{aligned}$$

36. We have,

$$\frac{(x-10)(x-20)}{(x-30)} < 0$$



Thus, $x \in (0, 10) \cup (20, 30)$

$$\text{Hence, the required probability} = \frac{18}{100} = \frac{9}{50}$$

37. We have,

$$\begin{aligned}
 x^2 - 6x^2 + 11x - 6 &= 0 \\
 \Rightarrow (x-1)(x-2)(x-3) &= 0 \\
 \Rightarrow x &= 1, 2, 3
 \end{aligned}$$

$$\text{Hence, the required probability} = \frac{3}{10}$$

38. (i) Hence, the required probability

$$\begin{aligned}
 &= \frac{{}^{50}C_1 \times {}^{33}C_1}{{}^{100}C_2} \\
 &= \frac{50 \times 33 \times 2}{100 \times 99} \\
 &= \frac{1}{3}
 \end{aligned}$$

(ii) Hence, the required probability

$$= \frac{{}^{50}C_2 + {}^{33}C_2 - {}^{16}C_2}{{}^{100}C_2}$$

(iii) Hence, the required probability = $\frac{{}^{16}C_2}{{}^{100}C_2}$

$$= \frac{4}{165}$$

(iv) Hence, the required probability = $1 - \frac{{}^{16}C_2}{{}^{100}C_2}$

$$\begin{aligned}
 &= 1 - \frac{4}{165} \\
 &= \frac{161}{165}
 \end{aligned}$$

(v) Hence, the required probability = $\frac{{}^{25}C_1 \times {}^{33}C_1}{{}^{100}C_2}$

$$\begin{aligned}
 &= \frac{25 \times 33 \times 2}{100 \times 99} \\
 &= \frac{1}{6}
 \end{aligned}$$

$$\begin{aligned}
 39. \text{ Hence, the required probability} &= \frac{7 \times 8 + 7 \times 8}{{}^{64}C_2} \\
 &= \frac{2 \times 112}{64 \times 63} \\
 &= \frac{2 \times 7}{4 \times 63} = \frac{7}{126} = \frac{1}{18}
 \end{aligned}$$

40. Hence, the required probability = $P(S \geq 5)$

$$\begin{aligned}
 &= 1 - P(S < 5) \\
 &= 1 - \{P(S=3) + P(S=4)\} \\
 &= 1 - \left\{ \frac{1}{216} + \frac{3}{216} \right\} \\
 &= \left(1 - \frac{4}{216} \right) = \left(1 - \frac{1}{54} \right) = \frac{53}{54}
 \end{aligned}$$

41. Hence, the required probability = $P(S \geq 15)$

$$\begin{aligned}
 &= P(S=15) + P(S=16) + P(S=17) + P(S=18) \\
 &= \frac{10+6+3+1}{216} = \frac{20}{216} = \frac{5}{54}
 \end{aligned}$$

42. Let S be the sample space.

$$\text{Then } n(S) = 2 \times 2 \times 2 \times 2 = 16$$

Let E be the event that the determinant is non-negative

Then E' be the event that the determinant is negative

$$= \left\{ \begin{vmatrix} 1 & 1 \\ 1 & 0 \end{vmatrix}, \begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix}, \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} \right\}$$

$$\text{Now, } P(E') = \frac{n(E')}{n(S)} = \frac{3}{16}$$

Hence, the required probability, $P(E)$

$$= 1 - P(E') = 1 - \frac{3}{16} = \frac{13}{16}$$

43. When $4 \rightarrow (1, 1, 2)$

$$P(S=4) = \frac{3!}{2!} = 3$$

$$16 \rightarrow (4, 6, 6), (5, 5, 6)$$

$$P(S=16) = \frac{3!}{2!} + \frac{3!}{2!} = 6$$

$$9 \rightarrow (1, 2, 6), (1, 3, 5), (2, 3, 4), (1, 4, 4)$$

$$(2, 2, 5), (3, 3, 3)$$

$$P(S=9) = 3 \times 3! + 2 \times \frac{3!}{2!} + 1 = 25$$

Hence, the required probability

$$\begin{aligned}
 &= P(S=4) + P(S=9) + P(S=16) \\
 &= \frac{3+6+25}{216} = \frac{34}{216} = \frac{17}{108}
 \end{aligned}$$

44. Let E be the event, where the product of two number is equal to the third number.

$$= \{(2, 5, 10), (2, 4, 8), (2, 3, 6)\}$$

$$\begin{aligned}\text{Hence, the required probability} &= \frac{3}{{}^{10}C_3} \\ &= \frac{3 \times 6}{10 \times 9 \times 8} = \frac{1}{40}\end{aligned}$$

45. Let n be the sides of a polygon.

$\therefore$ Each interior angle $= 150^\circ$

$$\Rightarrow \frac{(2n-4) \times 90}{n} = 150$$

$$\Rightarrow 180n - 360 = 150n$$

$$\Rightarrow 30n = 360$$

$$\Rightarrow n = 12$$

Hence, the required probability

$$\begin{aligned}&\frac{12(12-3)}{{}^{12}C_2} = \frac{12 \times 9}{12 \times 11} = \frac{9}{11}\end{aligned}$$

$$\begin{aligned}46. \text{ Hence, the required probability} &= \frac{{}^3P_3}{3^3} \\ &= \frac{3!}{3^3} = \frac{6}{27} = \frac{2}{9}\end{aligned}$$

47. Let E be the event, where the number of selections of two diagonals which intersect at an interior point.

Thus, $n(E)$ = the total number of selection of four vertices $= {}^6C_4$

$$\begin{aligned}\text{Hence, the required probability} &= \frac{{}^6C_4}{{}^9C_2} \\ &= \frac{{}^6C_2}{{}^9C_2} = \frac{6 \times 5}{9 \times 8} = \frac{5}{12}\end{aligned}$$

$$48. \text{ Hence, the required probability} = \frac{2}{{}^6C_3} = \frac{2}{5 \times 4} = \frac{1}{10}$$

$$49. \text{ Hence, the required probability} = \frac{{}^nP_n}{(n^n)} = \left(\frac{n!}{n^n} \right)$$

50. Hence, the required probability

$$= \frac{5! \times \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} \right)}{{}^5P_5}$$

$$= \frac{5! \times \left(\frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} \right)}{{}^5P_5}$$

$$= \frac{120 \times \left(\frac{1}{2} - \frac{1}{6} + \frac{1}{24} - \frac{1}{120} \right)}{120}$$

$$= \left(\frac{1}{2} - \frac{1}{6} + \frac{1}{24} - \frac{1}{120} \right)$$

$$= \frac{60 - 20 + 5 - 1}{120} = \frac{44}{120} = \frac{11}{30}$$

51. We have,

$$7 \rightarrow (1, 1, 1, 1, 3), (1, 1, 1, 2, 2)$$

Thus, the number of onto functions

$$\begin{aligned}&= \frac{7!}{3!} \times \frac{5!}{4!} + \frac{7!}{2! \times 2!} \times \frac{5!}{3! \times 2!} \\ &= \frac{7! \times 5!}{36}\end{aligned}$$

Hence, the required probability

$$= \frac{\left(\frac{7! \times 5!}{36} \right)}{5^7} = \frac{7! \times 5!}{5^7 \times 36} = \frac{7! \times 5 \times 24}{5^7 \times 36} = \frac{7! \times 2}{5^6 \times 3}$$

52. Of all the mapping that can be defined from the set $A = \{1, 2, 3, 4\}$ to the set $B = \{5, 6, 7, 8, 9\}$, a mapping is a selected at random. Find the probability that the selected mapping is strictly monotonic.

53. Hence, the required probability

$$= \frac{\frac{6!}{3! \times 3!}}{6 \times 6 \times 6} = \frac{20}{216} = \frac{5}{54}$$

54. Hence, the required probability

$$\begin{aligned}&= \frac{7 \times 7}{{}^{64}C_4} = \frac{49 \times 24}{64 \times 63 \times 62 \times 61} \\ &= \frac{7}{24 \times 62 \times 61} = \frac{7}{90768}\end{aligned}$$

55. Hence, the required probability

$$\left(1 - \frac{1}{3!} \right) = 1 - \frac{1}{6} = \frac{5}{6}$$

$$56. \text{ Hence, the required probability} = \frac{{}^{16}C_3}{{}^{100}C_3}$$

$$\begin{aligned}&= \frac{16 \times 15 \times 14}{100 \times 99 \times 98} \\ &= \frac{4}{1155}\end{aligned}$$

57. Here, $S = \{1, 2, 3, \dots, 100\}$.

Let A be the event, where the natural numbers is divisible by 2 $= \{2, 4, 6, \dots, 100\}$

and B be the event, where the natural numbers is divisible by both 3 $= \{3, 6, 9, \dots, 99\}$

and $A \cap B$ is the event, which is divisible by both 2 and 3 $= \{6, 12, 18, \dots, 96\}$

Hence, the required probability

$$\begin{aligned}&= P(A \text{ or } B) \\ &= P(A \cup B) \\ &= P(A) + P(B) - P(A \cap B) \\ &= \frac{50}{100} + \frac{33}{100} - \frac{16}{100} = \frac{67}{100}\end{aligned}$$

58. Here, $S = \{1, 2, 3, \dots, 200\}$.

Let A be the event, where the natural numbers is divisible by 3 = $\{3, 6, \dots, 198\}$.

B be the event, where the natural numbers is divisible by 4 = $\{4, 8, 12, \dots, 200\}$

and C be the event, where the natural numbers is divisible by 5 = $\{5, 10, 15, \dots, 200\}$

Hence, the required probability

$$\begin{aligned}
 &= P(A \text{ or } B \text{ or } C) \\
 &= P(A \cup B \cup C) \\
 &= P(A) + P(B) + P(C) - P(A \cap B) \\
 &\quad - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C) \\
 &= \frac{66}{200} + \frac{50}{200} + \frac{40}{200} - \frac{16}{200} - \frac{13}{200} - \frac{10}{200} + \frac{3}{200} \\
 &= \frac{120}{200} = \frac{3}{5}
 \end{aligned}$$

59. Hence, the required probability

$$\begin{aligned}
 &= P(C \text{ or } F) = P(C \cup F) \\
 &= P(C) + P(F) - P(C \cap F) \\
 &= \frac{{}^{16}C_1 + {}^{12}C_1 - {}^{12}C_1}{{}^{52}C_1 + {}^{52}C_1 - {}^{52}C_1} = \frac{16}{52} = \frac{4}{13}
 \end{aligned}$$

60. Required Probability

$$\begin{aligned}
 &= P(K \cup H \cup R) \\
 &= P(K) + P(H) + P(R) - P(K \cap H) \\
 &\quad - P(K \cap R) + P(H \cap R) + P(K \cap H \cap R) \\
 &= \frac{4}{52} + \frac{13}{52} + \frac{26}{52} - \frac{1}{52} - \frac{2}{52} - \frac{13}{52} + \frac{1}{52} \\
 &= \frac{28}{52} = \frac{7}{13}
 \end{aligned}$$

61. Here, $S = \{(1, 1), (1, 2), (1, 3), \dots, (6, 6)\}$

Let A be the event, where the sum of the numbers of two faces is divisible by 4

$$\begin{aligned}
 &= \{(1, 3), (3, 1), (2, 2), (2, 6), (6, 2) \\
 &\quad (3, 5), (5, 3), (4, 4), (6, 6)\}
 \end{aligned}$$

Let B be the event, where the sum of the numbers of two faces is divisible by 5.

$$= \{(1, 4), (4, 1), (2, 3), (3, 2), (4, 6), (6, 4), (5, 5)\}$$

Clearly A and B are mutually exclusive events.

Thus, the required probability

$$\begin{aligned}
 &= P(A \cup B) + P(A) + P(B) \\
 &= \frac{9}{36} + \frac{7}{36} = \frac{16}{36} = \frac{4}{9}
 \end{aligned}$$

62. Here, $S = \{(1, 1), (1, 2), (1, 3), \dots, (6, 6)\}$

Let A be the event, where the first die always shows 3

$$= \{(3, 1), (3, 2), \dots, (3, 6)\}$$

B be the event where the second die always shows 3

$$= \{(1, 3), (2, 3), \dots, (6, 3)\}$$

and $A \cap B$ be the event, where both the die always shows 3 = $\{(3, 3)\}$

Hence, the required probability = $P(A \cup B)$

$$\begin{aligned}
 &= P(A) + P(B) - P(A \cap B) \\
 &= \frac{6}{36} + \frac{6}{36} - \frac{1}{36} = \frac{11}{36}
 \end{aligned}$$

63. Here, $S = \{(1, 1), (1, 2), (1, 3), \dots, (6, 6)\}$.

Let A be the event, where the first die always shows odd numbers = $\{(1, 1), (1, 2), \dots, (1, 6), (3, 1), \dots, (3, 6), (5, 1), \dots, (5, 6)\}$

Let B be the event, where the sum of two outcome is 7.

$$= \{(1, 6), (6, 1), (2, 5), (5, 2), (3, 4), (4, 3)\}$$

and $A \cap B$ be the event, where first die is an odd number and sum of the outcome is a 7

$$= \{(1, 6), (3, 4), (5, 2)\}$$

Hence, the required probability = $P(A \cup B)$

$$= P(A) + P(B) - P(A \cap B)$$

$$= \frac{18}{36} + \frac{6}{36} - \frac{3}{36} = \frac{21}{36} = \frac{7}{12}$$

64. Here, $S = \{(1, 1), (1, 2), (1, 3), \dots, (6, 6)\}$.

Let A be the event, where the sum of the numbers is coming up more than 9

$$= \{(4, 6), (6, 4), (5, 5), (5, 6), (6, 5), (6, 6)\}$$

Hence, the required probability = $\frac{6}{36} = \frac{1}{6}$

65. Hence the required probability = $P(I \cup A)$

$$= P(I) + P(A) - P(I \cap A)$$

$$= \frac{1}{3} + \frac{1}{2} - \frac{1}{6} = \frac{4}{6} = \frac{2}{3}$$

66. Required Probability = $P(S \cup C \cup H \cup D)$

$$= P(S) + P(C) + P(H) + P(D)$$

$$= \frac{{}^{13}C_4}{{}^{52}C_4} + \frac{{}^{13}C_4}{{}^{52}C_4} + \frac{{}^{13}C_4}{{}^{52}C_4} + \frac{{}^{13}C_4}{{}^{52}C_4}$$

$$= \frac{4 \times {}^{13}C_4}{{}^{52}C_4}$$

67. Hence, the required probability = $P(M \cup P)$

$$= P(M) + P(P) - P(P \cap M)$$

$$= \frac{1}{3} + \frac{1}{4} - \frac{1}{5} = \frac{20 + 15 - 12}{60} = \frac{23}{60}$$

68. Hence, the required probability

$$= \frac{{}^{26}C_2 + {}^{26}C_2}{{}^{52}C_2}$$

$$= \frac{2 \times {}^{26}C_2}{{}^{52}C_2}$$

$$= \frac{2 \times 26 \times 25}{52 \times 51}$$

$$= \frac{25}{51}$$

69. $P(E \cup P) = P(E) + P(P) - P(E \cap P)$

$$\Rightarrow \frac{2}{3} = \frac{2}{5} + \frac{4}{7} - P(E \cap P)$$

$$\Rightarrow P(E \cap P) = \frac{2}{5} + \frac{4}{7} - \frac{2}{3}$$

$$\Rightarrow P(E \cap P) = \frac{32}{105}$$

70. Hence, the required probability = $P(M \cup E)$

$$= P(M) + P(E) - P(M \cap E)$$

$$= \frac{2}{3} + \frac{4}{9} - \frac{1}{4}$$

$$= \frac{31}{36}$$

71. Hence, the required probability = $P(P \cup T)$

$$= P(P) + P(T)$$

$$= \frac{3}{5} + \frac{1}{4}$$

$$= \frac{17}{20}$$

72. Hence, the required probability

$$= P(k \text{ or } H \text{ or } R)$$

$$= P(k \cup H \cup R)$$

$$= P(k) + P(H) + P(R) - P(K \cap H)$$

$$\quad - P(K \cap R) - P(R \cap H) - P(K \cap H \cap R)$$

$$= \frac{4}{52} + \frac{13}{52} + \frac{26}{52} - \frac{1}{52} - \frac{2}{52} - \frac{13}{52} + \frac{1}{52}$$

$$= \frac{4}{52} + \frac{26}{52} - \frac{2}{52}$$

$$= \frac{28}{52} = \frac{7}{13}$$

73. Hence, the required probability

$$= \frac{{}^{13}C_2 + {}^{13}C_2 + {}^{13}C_2}{{}^{52}C_2}$$

$$= \frac{3 \times {}^{13}C_2}{{}^{52}C_2}$$

$$= \frac{3 \times 13 \times 12}{52 \times 51} = \frac{3}{17}$$

74. Let A , B and C be the events represents integers which are multiple of 2, 3 and 10.

Hence, the required probability

$$= P(A \cup B \cup C)$$

$$= P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C)$$

$$\quad - P(B \cap C) + P(A \cap B \cap C)$$

$$= \frac{25}{50} + \frac{16}{50} + \frac{5}{50} - \frac{8}{50} - \frac{5}{50} - \frac{1}{50} + \frac{1}{50}$$

$$= \frac{25}{50} + \frac{16}{50} - \frac{8}{50}$$

$$= \frac{34}{50}$$

75. $4 \rightarrow (1, 3), (3, 1), (2, 2)$

$7 \rightarrow (1, 6), (6, 1), (2, 5), (5, 2), (4, 3), (3, 4)$

$12 \rightarrow (6, 6)$

Hence, the required probability

$$= \frac{10}{36} = \frac{5}{18}$$

76. $P((A \cap \bar{B}) \cup (\bar{A} \cap B)) = 1 - x$... (i)

$P((B \cap \bar{C}) \cup (\bar{B} \cap C)) = 1 - 2x$... (ii)

$P((C \cap \bar{A}) \cup (\bar{C} \cap A)) = 1 - x$... (iii)

Adding (i), (ii) and (iii), we get,

$$2(P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C))$$

$$= 3 - 4x$$

Now, $P(A \cup B \cup C)$

$$= P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C)$$

$$\quad - P(C \cap A) + P(A \cap B \cap C)$$

$$= \frac{3-4x}{2} + x^2$$

$$= \frac{2x^2 - 4x + 3}{2}$$

$$= \frac{2(x^2 - 2x + 1) + 1}{2}$$

$$= (x-1)^2 + \frac{1}{2} \geq \frac{1}{2}$$

Hence, the result.

77. (i) From the definition of in-equalities of probab we can write

$$P(A \cup B) \leq P(B) \Rightarrow P(A \cup B) \geq \frac{2}{3}$$

(ii) From the definition of in-equalities of probab we can write

$$P(A \cap B) \leq P(A)$$

$$\Rightarrow P(A \cap B) \leq \frac{3}{5}$$

Also,

$$P(A \cap B) = P(A) + P(B) - P(A \cup B)$$

$$\geq P(A) + P(B) - 1$$

$$\Rightarrow P(A \cap B) \geq \frac{3}{5} + \frac{2}{3} - 1 = \frac{9+10-15}{15} = \frac{4}{15}$$

Hence from the above two relation, we get

$$\frac{4}{15} \leq P(A \cap B) \leq \frac{3}{5}$$

78.

(i) From the definition of in-equalities of probab we can write

$$P(A \cup B) \geq P(A) + P(B) - 1$$

$$= \frac{3}{4} + \frac{3}{8} - 1 = \frac{6+3-8}{8} = \frac{1}{8}$$

(ii) From the definition of in-equalities of probab we can write

$$P(A) + P(B) - 1 \leq P(A \cap B) \leq P(A)$$

$$\Rightarrow \frac{1}{8} \leq P(A \cap B) \leq \frac{3}{4}$$

79. Let $S = \{1, 2, 3, 4, 5, 6\}$ and $B = \{1, 3, 5\}$ and $A = \{2, 3, 5\}$.

$$\text{Required Probability} = P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = \frac{2}{3}$$

80. Let $S = \{(1, 1), (1, 2), \dots, (6, 6)\}$

$B = \{(1, 2), \dots, (6, 2), (1, 3),$

$\dots, (6, 3), (1, 5), \dots, (6,$

and $A = \{(2, 6), (6, 2), (3, 5), (5, 3), (4, 4)\}$.

$$\text{Required Probability} = \frac{P(A \cap B)}{P(B)} = \frac{3}{5}$$

81. Here, $S = \{(1, 1), (1, 2), \dots, (6, 6)\}$.

Let A be the event, the sum of the numbers coming up is 9 = $\{(3, 6), (6, 3), (5, 4), (4, 5)\}$ and

B be the event, where first will shows 5

$$= \{(5, 1), (5, 3), (5, 3), \dots, (5, 6)\}$$

and $A \cap B$ be the event, where sum of two outcomes is 9 and first die shows 5 = $\{(5, 4)\}$

Hence, the required probability

$$= P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = \frac{1}{6}$$

82. Here, $S = \{(1, 1), (1, 2), \dots, (6, 6)\}$.

Let A be the event of coming up two heads = $\{(H, H)\}$

and B be the event of coming up at least one head

$$= \{(H, H), (H, T), (T, H)\}$$

Hence, the required probability = $P\left(\frac{A}{B}\right)$

$$= \frac{P(A \cap B)}{P(B)} = \frac{1}{3}$$

83. Here, $S = \{(1, 1), (1, 2), \dots, (6, 6)\}$.

Let A be the event, the sum of the numbers coming up is 8

$$= \{(3, 5), (5, 3), (4, 4), (2, 6), (6, 2)\}$$

B be the event, where first will shows 4

$$= \{(4, 1), (4, 2), (4, 3), \dots, (4, 6)\}$$

and $A \cap B$ be the event, where the sum of two outcomes is 8 and first die shows 4 = $\{(4, 4)\}$

Hence, the required probability = $P\left(\frac{A}{B}\right)$

$$= \frac{P(A \cap B)}{P(B)} = \frac{1}{6}$$

84. Here, $S = \{(1, 1), (1, 2), \dots, (6, 6)\}$

Let A be the event, the sum of the numbers coming up is 7 = $\{(1, 6), (6, 1), (2, 5), (5, 2), (3, 4), (4, 3)\}$

B be the event, where second die will show a prime number.

$$= \{(1, 2), \dots, (6, 2), (1, 3), \dots, (6, 3), (1, 5), \dots, (6, 5)\}$$

and $A \cap B$ be the event, where sum of two outcomes is 7 and second die always exhibits a prime number

$$= \{(2, 2), (2, 5), (4, 3)\}$$

Hence, the required probability

$$= P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = \frac{3}{18} = \frac{1}{6}$$

85. Here, $S = \{(1, 1), (1, 2), \dots, (6, 6)\}$.

Let A be the event, the sum of the numbers coming up is 7 = $\{(1, 6), (6, 1), (2, 5), (5, 2), (3, 4), (4, 3)\}$

B be the event, where second die always exhibits an odd number = $\{(1, 1), \dots, (6, 1), (1, 3), \dots, (6, 3), (1, 5), \dots, (6, 5)\}$

and $A \cap B$ be the event, where sum of two outcomes is 7 and second die always exhibits an odd number = $\{(6, 1), (2, 5), (4, 3)\}$

Hence, the required probability

$$= P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = \frac{3}{18} = \frac{1}{6}$$

86. Here, $S = \{(1, 1), (1, 2), \dots, (6, 6)\}$

Let A be the event, the sum of the numbers coming up is 8 = $\{(3, 5), (5, 3), (4, 4), (2, 6), (6, 2)\}$

B be the event, where second die will show 4

$$= \{(1, 4), (2, 4), \dots, (6, 4)\}$$

and $A \cap B$ be the event, where sum of two outcomes is 8 and second die shows 4 = $\{(4, 4)\}$

Hence, the required probability

$$= P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = \frac{1}{6}$$

87. Here, $S = \{(1, 1), (1, 2), \dots, (6, 6)\}$.

Let A be the event, the sum of the numbers coming up is 6 = $\{(1, 5), (5, 1), (2, 4), (4, 2), (3, 3)\}$

B be the event, where 4 has appeared at least once.

$$= \{(1, 4), (2, 4), (4, 4), \dots, (6, 4), (4, 1), \dots, (4, 6)\}$$

and $A \cap B$ be the event, where the sum of two outcomes is 6 and 4 has appeared at least once = $\{(2, 4), (4, 2)\}$

Hence, the required probability

$$= P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = \frac{2}{11}$$

88. To test the quality of electric bulbs produced in a factory, two bulbs are randomly selected from a large sample without replacement. If either bulb is defective, the entire lot is rejected. Suppose a sample of 200 bulbs contains 5 defective bulbs. Find the probability that the sample will be rejected.

$$\begin{aligned} 89. \text{ Hence, the required probability} &= \frac{10}{25} \times \frac{15}{24} \\ &= \frac{2}{5} \times \frac{15}{24} \\ &= \frac{3}{12} = \frac{1}{4} \end{aligned}$$

$$\begin{aligned} 90. \text{ Hence, the required probability} &= \frac{13}{52} \times \frac{12}{51} \\ &= \frac{1}{4} \times \frac{12}{51} \\ &= \frac{3}{51} = \frac{1}{17} \end{aligned}$$

$$\begin{aligned} 91. \text{ Hence, the required probability} &= \frac{{}^5C_4}{{}^{20}C_4} \\ &= \frac{5.4.3.2}{20.19.18.17} \\ &= \frac{1}{19.3.17} \end{aligned}$$

$$= \frac{1}{19 \times 51}$$

$$= \frac{1}{9 \cdot 69}$$

9.2 Let $S = \{1, 2, 3, \dots, 11\}$.

Hence, the required probability = $\frac{{}^6C_2}{{}^6C_2 \times {}^5C_2}$

$$= \frac{6.5}{6.5 + 5.4}$$

$$= \frac{30}{30 + 20}$$

$$= \frac{3}{5}$$

9.3 Let $S = \{B_1B_2, B_1G_2, G_1B_2, G_1G_2\}$.

(i) Let A be the event, where one children is a boy,
i.e. $A = \{B_1B_2, B_1G_2, G_1B_2\}$

Hence, the required probability = $\frac{1}{3}$

(ii) Let B be the event, where older child is a boy,
i.e. $B = \{B_1G_2, B_1B_2\}$

Hence, the required probability = $\frac{1}{2}$

94. Let $S = \{HH, HT, TH, TT\}$

and $A = \{HH, GT, TH\}, B = \{HH\}$

Hence, the required probability = $\frac{1}{3}$

9.5 Hence, the required probability = $\frac{2}{26} = \frac{1}{13}$

95. Let A be the event of drawing a king card and B be the event of drawing a red card.

Hence, the required probability

$$= P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = \frac{{}^{26}C_1}{{}^{26}C_1} = \frac{2}{26} = \frac{1}{13}$$

9.6 We have, Bag 1 $\rightarrow (4W, 2B_1)$

and Bag 2 $\rightarrow (3W, 5B_2)$

Here, W_1 and W_2 , and B_1 and B_2 are independents.

(i) $P(W_1 \cap W_2) = P(W_1) \cdot P(W_2) = \frac{4}{6} \times \frac{3}{8} = \frac{1}{4}$

(ii) $P(B_1 \cap B_2) = P(B_1) \cdot P(B_2) = \frac{1}{3} \times \frac{5}{8} = \frac{5}{24}$

(iii) $P\{(W_1 \cap B_2) \cup (W_2 \cap B_1)\}$
 $= P(W_1 \cap B_2) + P(W_2 \cap B_1)$
 $= P(W_1) \cdot P(B_2) + P(W_2) \cdot P(B_1)$
 $= \frac{2}{3} \times \frac{5}{8} + \frac{1}{3} \times \frac{3}{8} = \frac{5}{12} + \frac{1}{8} = \frac{13}{24}$

9.7 Let three students are A, B, C .

Consider the problem is not solved by any one of them.

Thus,

$$P(\bar{A} \cap \bar{B} \cap \bar{C}) = P(\bar{A}) \cdot P(\bar{B}) \cdot P(\bar{C})$$

$$= \left(1 - \frac{1}{2}\right) \cdot \left(1 - \frac{1}{3}\right) \cdot \left(1 - \frac{1}{4}\right)$$

$$= \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} = \frac{1}{4}$$

Hence, the required probability is $= 1 - \frac{1}{4} = \frac{3}{4}$

9.8 Hence, the required probability

$$= P(A \cap \bar{B} \text{ or } \bar{A} \cap B)$$

$$= P(A \cap \bar{B}) + P(\bar{A} \cap B)$$

$$= P(A) \cdot P(\bar{B}) + P(\bar{A}) \cdot P(B)$$

$$= \frac{75}{100} \times \frac{20}{100} + \frac{25}{100} \times \frac{80}{100}$$

$$= \frac{15}{100} + \frac{20}{100} = \frac{7}{20}$$

$$= \frac{35}{100}$$

$$= 35\%$$

9.9 Let three students are A, B, C respectively.

Hence, the required probability

$$= P(A \cap B \cap \bar{C}) + P(A \cap \bar{B} \cap C)$$

$$+ P(\bar{A} \cap B \cap C) + P(A \cap B \cap C)$$

$$= P(A) \cdot P(B) \cdot P(\bar{C})$$

$$+ P(A) \cdot P(\bar{B}) \cdot P(C) + P(\bar{A}) \cdot P(B) \cdot P(C)$$

$$+ P(A) \cdot P(B) \cdot P(C)$$

$$= \frac{1}{3} \cdot \frac{1}{4} \cdot \left(1 - \frac{1}{5}\right) + \frac{1}{3} \cdot \left(1 - \frac{1}{4}\right) \cdot \frac{1}{5}$$

$$+ \left(1 - \frac{1}{3}\right) \cdot \frac{1}{4} \cdot \frac{1}{5} + \frac{1}{3} \cdot \frac{1}{4} \cdot \frac{1}{5}$$

$$= \frac{4}{60} + \frac{2}{60} + \frac{3}{60} + \frac{1}{60} = \frac{10}{60} = \frac{1}{6}$$

100. Let two persons be A and B , respectively. Consider none of them hit the target.

Now, $= P(A \cap B \cap \bar{C}) + P(A \cap \bar{B} \cap C)$

$$= \left(1 - \frac{3}{4}\right) \times \left(1 - \frac{2}{3}\right) = \frac{1}{12}$$

Hence, the required probability $= 1 - \frac{1}{12} = \frac{11}{12}$

101. Let E_1 be the of throwing 6 with a pair of dice and E_2 be the event of throwing 7 with a pair of dice.

Thus, $E_1 = \{(1, 5), (5, 1), (2, 4), (4, 2), (3, 3)\}$

and

$$E_2 = \{(1, 6), (6, 1), (2, 5), (5, 2), (3, 4), (4, 3)\}$$

The probability of A can win

$$\begin{aligned}
 P(A) &= P(E_1 \text{ or } \overline{E_1} \cdot \overline{E_2} \cdot E_1 \text{ or } \overline{E_1} \cdot \overline{E_2} \cdot \overline{E_1} \cdot \overline{E_2} \cdot E_1 \text{ or } \dots) \\
 &= \frac{5}{36} + \frac{31}{36} \cdot \frac{5}{6} \cdot \frac{5}{36} + \left(\frac{31}{36}\right)^2 \left(\frac{5}{6}\right)^2 \left(\frac{5}{36}\right) + \dots \infty \\
 &= \frac{5}{36} + \left(\frac{155}{216}\right) \cdot \frac{5}{36} + \left(\frac{155}{216}\right)^2 \left(\frac{5}{36}\right) + \dots \infty \\
 &= \frac{\frac{5}{36}}{1 - \frac{155}{216}} = \frac{30}{61}
 \end{aligned}$$

The probability that B can win = $1 - \frac{30}{61} = \frac{31}{61}$.

102. Let A be the event that a randomly selected ball is a red. E_1 and E_2 be denote two bags respectively.

Then $P(A) = P(E_1) P(A/E_1) + P(E_2) P(A/E_2)$

$$= \frac{1}{2} \times \frac{4}{7} + \frac{1}{2} \times \frac{2}{6} = \frac{2}{7} + \frac{1}{6} = \frac{19}{42}$$

103. **Case I:** When a white ball is transferred from the first bag to a second bag.

Then $P(W) = \frac{4}{9} \times \frac{7}{14} = \frac{2}{9}$

Case II: When a black ball is transferred from the first bag to a 2nd bag.

Then $P(W) = \frac{4}{9} \times \frac{6}{14} = \frac{2}{3} \times \frac{2}{7} = \frac{4}{21}$

Hence, the required probability = $\frac{2}{9} + \frac{4}{21} = \frac{14+12}{63} = \frac{26}{63}$.

104. Let E_1 , E_2 and A denote the events, the lot contains 2 defective, 3 defective and all the defective articles are found by 12th test, i.e. the event A means that all but one defective article must be found in the first eleven testings.

Now, $P(E_1) = 0.4$, $P(E_2) = 0.6$

$$P(A/E_1) = \frac{{}^2C_1 \times {}^{18}C_{10}}{{}^{20}C_{11}} \times \frac{1}{9} = \frac{11}{190}$$

$$\text{and } P(A/E_2) = \frac{{}^3C_2 \times {}^{17}C_9}{{}^{20}C_{11}} \times \frac{1}{9} = \frac{11}{228}$$

Hence the required probability

$$\begin{aligned}
 &= P(A) \\
 &= P(E_1) P(A/E_1) + P(E_2) P(A/E_2) \\
 &= (0.4) \left(\frac{11}{190}\right) + (0.6) \left(\frac{11}{228}\right) = \frac{99}{1900}
 \end{aligned}$$

105. Let E_{xy} denote the event that the first ball drawn has color x and second ball drawn has color y and A be the event that the third ball drawn is black.

$$P(E_{ww}) = \frac{2}{4} \times \frac{1}{3} = \frac{1}{6}, P(E_{wb}) = \frac{2}{4} \times \frac{2}{3} = \frac{1}{3},$$

$$P(E_{bw}) = \frac{2}{4} \times \frac{2}{5} = \frac{1}{5}, P(E_{bb}) = \frac{2}{4} \times \frac{3}{5} = \frac{3}{10}$$

Also, $P(A/E_{ww}) = \frac{2}{2} = 1$, $P(A/E_{wb}) = \frac{3}{4}$,

$$P(A/E_{bw}) = \frac{3}{4}, P(A/E_{bb}) = \frac{4}{6} = \frac{2}{3}$$

Hence the required probability

$$\begin{aligned}
 &= P(A) \\
 &= P(E_{ww}) \cdot P(A/E_{ww}) + P(E_{wb}) \cdot P(A/E_{wb}) \\
 &\quad + P(E_{bw}) \cdot P(A/E_{bw}) + P(E_{bb}) \cdot P(A/E_{bb}) \\
 &= \frac{1}{6} \times 1 + \frac{1}{3} \times \frac{3}{4} + \frac{1}{5} \times \frac{3}{4} + \frac{3}{10} \times \frac{2}{3} = \frac{23}{30}
 \end{aligned}$$

106. Let E_1 , E_2 denote the events that the coin shows a head, tail and A be the event that the noted number is either 7 or 8.

We have $P(E_1) = \frac{1}{2}$ and $P(E_2) = \frac{1}{2}$

Now, $7 \rightarrow \{(1, 6), (6, 1), (2, 5), (5, 2), (3, 4), (4, 3)\}$

and $8 \rightarrow \{(2, 6), (6, 2), (3, 5), (5, 3), (4, 4)\}$

Thus, $P(A/E_1) = \frac{11}{36}$, $P(A/E_2) = \frac{2}{11}$

Hence, the required probability

$$\begin{aligned}
 &= P(A) = P(E_1) P(A/E_1) + P(E_2) P(A/E_2) \\
 &= \left(\frac{1}{2}\right) \left(\frac{11}{36}\right) + \left(\frac{1}{2}\right) \left(\frac{2}{11}\right) = \frac{193}{792}
 \end{aligned}$$

107. Let W be the event that a white ball is drawn at the first draw B be the event that a black ball is drawn at the first draw and A be the event that a white ball is drawn at the second draw.

We have, $P(W) = \frac{m}{m+n}$, $P(B) = \frac{n}{m+n}$

and $P(A/W) = \frac{m+k}{m+n+k}$, $P(A/B) = \frac{m}{m+n+k}$

Hence the required probability,

$$\begin{aligned}
 P(A) &= P(W) \times P(A/W) + P(B) \cdot P(A/B) \\
 &= \frac{m}{m+n} \cdot \frac{m+k}{m+n+k} + \frac{n}{m+n} \cdot \frac{m}{m+n+k} \\
 &= \frac{m}{m+n}
 \end{aligned}$$

108. Hence the required probability,

$$\begin{aligned}
 P(E_2/B) &= \frac{P(E_2) P(A/E_2)}{P(E_1) P(A/E_1) + P(E_2) P(A/E_2)} \\
 &= \frac{\frac{1}{2} \times \frac{5}{11}}{\frac{1}{2} \times \frac{3}{7} + \frac{1}{2} \times \frac{5}{11}} = \frac{35}{68}
 \end{aligned}$$

109. It is given that

$$P(A) = \frac{4}{5}, P(B) = \frac{3}{4}, P(C) = \frac{2}{3}$$

(i) Hence, the required probability,

$$\begin{aligned} P(ABC) &= P(A)P(B)P(C) \\ &= \frac{4}{5} \times \frac{3}{4} \times \frac{2}{3} \\ &= \frac{2}{5} \end{aligned}$$

(ii) Hence, the required probability,

$$\begin{aligned} P(\bar{A}\bar{B}C) &= P(\bar{A})P(B)P(C) \\ &= \left(1 - \frac{4}{5}\right) \times \frac{3}{4} \times \frac{2}{3} \\ &= \frac{1}{5} \times \frac{3}{4} \times \frac{2}{3} \\ &= \frac{1}{10} \end{aligned}$$

110. Let A be the event that the man reports that six occurs on throwing a die. E_1 is the event that six occurs and E_2 is the event that six does not occur.

$$\begin{aligned} P(E_1) &= \frac{1}{6} \text{ and } P(E_2) = \frac{5}{6} \\ P(A/E_1) &= \frac{3}{5} \text{ and } P(A/E_2) = \frac{2}{5} \end{aligned}$$

Hence the required probability

$$\begin{aligned} P(E_1/A) &= \frac{P(E_1)P(A/E_1)}{P(E_1)P(A/E_1) + P(E_2)P(A/E_2)} \\ &= \frac{\frac{1}{6} \times \frac{3}{5}}{\frac{1}{6} \times \frac{3}{5} + \frac{5}{6} \times \frac{2}{5}} = \frac{3}{3+10} = \frac{3}{13} \end{aligned}$$

111. Hence, the required probability

$$\begin{aligned} &= \frac{{}^2C_1 \times {}^2C_1}{{}^{52}C_2} \\ &= \frac{2}{663} \end{aligned}$$

112. Hence, the required probability

$$\begin{aligned} &= \frac{{}^{13}C_1 \times {}^2C_1}{{}^{52}C_2} \\ &= \frac{13 \times 2 \times 2}{52 \times 51} \\ &= \frac{1}{51} \end{aligned}$$

113. Hence, the required probability

$$\begin{aligned} &= \frac{{}^3C_2 \times {}^2C_2}{{}^{25}C_2} \\ &= \frac{3 \times 2 \times 2}{25 \times 24} \\ &= \frac{1}{50} \end{aligned}$$

114. Hence, the required probability

$$\begin{aligned} &= 1 - \text{None of them solve the problem} \\ &= 1 - P(\bar{A}\bar{B}\bar{C}) \\ &= 1 - \{P(\bar{A}) \cdot P(\bar{B}) \cdot P(\bar{C})\} \\ &= 1 - \left\{\frac{1}{3} \times \frac{2}{7} \times \frac{3}{8}\right\} \\ &= 1 - \frac{1}{28} \\ &= \frac{27}{28} \end{aligned}$$

115. (i) Hence, the required probability,

$$\begin{aligned} P(A\bar{B}) &= P(A)P(\bar{B}) \\ &= \frac{3}{7} \times \left(1 - \frac{5}{7}\right) \\ &= \frac{3}{7} \times \frac{2}{7} \\ &= \frac{6}{49} \end{aligned}$$

(ii) Hence, the required probability,

$$\begin{aligned} P(A\bar{B} \text{ or } \bar{A}B) &= P(A\bar{B}) + P(\bar{A}B) \\ &= P(A)P(\bar{B}) + P(\bar{A})P(B) \\ &= \frac{3}{7} \times \frac{2}{7} + \frac{4}{7} \times \frac{5}{7} \\ &= \frac{26}{49} \end{aligned}$$

$$116. P(W) = \frac{1}{6}, P(F) = \frac{5}{6}$$

Hence, the required probability of A ,

$$\begin{aligned} P(A) &= P(W \text{ or } FFW \text{ or } FFFFW \text{ or } \dots \text{ to } \infty) \\ &= P(W) + P(FFW) + P(FFFFW) + \dots \\ &= \frac{1}{6} + \left(\frac{5}{6}\right)^2 \left(\frac{1}{6}\right) + \left(\frac{5}{6}\right)^4 \left(\frac{1}{6}\right) + \dots \\ &= \frac{1}{6} \left(1 + \left(\frac{5}{6}\right)^2 + \left(\frac{5}{6}\right)^4 + \dots\right) \\ &= \frac{1}{6} \times \frac{1}{1 - \left(\frac{5}{6}\right)^2} \end{aligned}$$

$$= \frac{1}{6} \times \frac{36}{36-25}$$

$$= \frac{6}{11}$$

Thus, $P(B) = 1 - P(A) = 1 - \frac{6}{11} = \frac{5}{11}$

117. Here, $6 \rightarrow (1, 5), (5, 1), (2, 4), (4, 2), (3, 3)$
and $7 \rightarrow (1, 6), (6, 1), (2, 5), (5, 2), (3, 4), (4, 3)$

$\therefore$ The probability of getting 6 is $\frac{5}{36}$

and the probability of getting 7 is $\frac{1}{6}$.

Thus, the probability of A

$$= \frac{5}{36} + \left(\frac{31}{36} \times \frac{5}{6} \right) \left(\frac{5}{36} \right) + \left(\frac{31}{36} \times \frac{5}{6} \right)^2 \left(\frac{5}{36} \right) + \dots$$

$$= \frac{5}{36} \left(1 + \left(\frac{31 \times 5}{36 \times 6} \right) + \left(\frac{31 \times 5}{36 \times 6} \right)^2 + \dots \right)$$

$$= \frac{5}{36} \times \frac{1}{1 - \left(\frac{31 \times 5}{36 \times 6} \right)}$$

$$= \frac{5}{36} \times \frac{36 \times 6}{216 - 155}$$

$$= \frac{30}{61}$$

Thus, the probability of B ,

$$P(B) = 1 - \frac{30}{61} = \frac{31}{61}$$

118. Here, $P(W) = \frac{1}{6}$, $P(F) = \frac{5}{6}$

Thus, the probability of A winning,

$$P(A) = P(W \text{ or } FFFW \text{ or } FFFFFFFW \text{ or } \dots)$$

$$= P(W) + P(FFFW) + P(FFFFFFFW) + \dots$$

$$= \frac{1}{6} + \left(\frac{5}{6} \right)^3 \frac{1}{6} + \left(\frac{5}{6} \right)^6 \frac{1}{6} + \dots$$

$$= \frac{1}{6} \left(1 + \left(\frac{5}{6} \right)^3 + \left(\frac{5}{6} \right)^6 + \dots \right)$$

$$= \frac{1}{6} \times \frac{1}{1 - \left(\frac{5}{6} \right)^3}$$

$$= \frac{1}{6} \times \frac{216}{216 - 125}$$

$$= \frac{36}{91}$$

Now, the probability of B winning

$$P(B) = P(FW \text{ or } FFFFW \text{ or } FFFFFFFW \text{ or } \dots)$$

$$= P(FW) + P(FFFW) + P(FFFFFFFW) + \dots$$

$$= \frac{5}{6} \times \frac{1}{6} + \left(\frac{5}{6} \right)^4 \times \frac{1}{6} + \left(\frac{5}{6} \right)^7 \times \frac{1}{6} + \dots$$

$$= \frac{5}{6} \times \frac{1}{6} \left(1 + \left(\frac{5}{6} \right)^3 + \left(\frac{5}{6} \right)^6 + \dots \right)$$

$$= \frac{5}{6} \times \frac{1}{6} \times \frac{1}{1 - \left(\frac{5}{6} \right)^3}$$

$$= \frac{5}{6} \times \frac{1}{6} \times \frac{216}{216 - 125}$$

$$= \frac{30}{91}$$

Hence, the probability of C winning

$$P(B) = 1 - \{P(A) + P(B)\}$$

$$= 1 - \left(\frac{36}{91} + \frac{30}{91} \right)$$

$$= 1 - \frac{66}{91}$$

$$= \frac{25}{91}$$

119. Hence, the required probability

$$= \frac{1}{2} \times \frac{4}{7} + \frac{1}{2} \times \frac{2}{6}$$

$$= \frac{1}{2} \times \left(\frac{4}{7} + \frac{1}{3} \right)$$

$$= \frac{1}{2} \times \frac{12+7}{21}$$

$$= \frac{19}{42}$$

120. Hence, the required probability $= \frac{4}{9} \times \frac{7}{14} + \frac{5}{9} \times \frac{6}{14}$
- $$= \frac{28+30}{126}$$
- $$= \frac{58}{126}$$
- $$= \frac{29}{63}$$

121. Let E_1 , E_2 and D denote the events, a lot contains 2 defective articles, second lot contains 3 defective articles and all the defective article are found by the 12th test, i.e. the event D means one defective articles must be found in the first eleven testings.

Now, $P(E_1) = 0.4$, $P(E_2) = 0.6$

$$P(D/E_1) = \frac{{}^2C_1 \times {}^{18}C_{10}}{{}^{20}C_{11}} \times \frac{1}{9} = \frac{11}{190}$$

$$P(D/E_2) = \frac{{}^3C_2 \times {}^{17}C_9}{{}^{20}C_{11}} \times \frac{1}{9} = \frac{11}{228}$$

Hence, the required probability,

$$\begin{aligned} P(D) &= P(E_1) \cdot P(D/E_1) + P(E_2) \cdot P(D/E_2) \\ &= 0.4 \times \frac{11}{190} + 0.6 \times \frac{11}{228} \\ &= \frac{99}{1900} \end{aligned}$$

122. Hence, the required probability

$$\begin{aligned} &= \frac{25}{100} \times \frac{5}{100} + \frac{35}{100} \times \frac{4}{100} + \frac{40}{100} \times \frac{2}{100} \\ &= \frac{125 + 140 + 80}{10000} \\ &= \frac{345}{10000} \\ &= \frac{69}{2000} \end{aligned}$$

123. Hence, the required probability

$$\begin{aligned} &= \frac{1}{3} \times \frac{25}{100} + \frac{2}{3} \times \frac{28}{100} \\ &= \frac{25 + 56}{300} \\ &= \frac{81}{300} \\ &= \frac{27}{100} \\ &= 27\% \end{aligned}$$

124. For the first two draw, Let the balls taken out may be

E_1 = white and white

E_2 = white and black

E_3 = black and white

E_4 = black and black

$$\text{Now, } P(E_1) = \frac{2}{4} \cdot \frac{1}{3} = \frac{1}{6}$$

$$P(E_2) = \frac{2}{4} \cdot \frac{2}{3} = \frac{1}{3}$$

$$P(E_3) = \frac{2}{4} \cdot \frac{2}{5} = \frac{1}{5}$$

$$P(E_4) = \frac{2}{4} \cdot \frac{3}{5} = \frac{3}{10}$$

Now,

$$P(E_1) + P(E_2) + P(E_3) + P(E_4)$$

$$= \frac{1}{6} + \frac{1}{3} + \frac{1}{5} + \frac{3}{10}$$

$$= \frac{10 + 20 + 12 + 18}{60}$$

$$= \frac{60}{60} = 1$$

Thus, the events E_1, E_2, E_3 and E_4 are exhaustive and hence they are mutually exclusive.

Hence, the required probability,

$$\begin{aligned} P(E) &= P(E_1)P\left(\frac{E}{E_1}\right) + P(E_2)P\left(\frac{E}{E_2}\right) \\ &\quad + P(E_3)P\left(\frac{E}{E_3}\right) + P(E_4)P\left(\frac{E}{E_4}\right) \\ &= \frac{1}{6} \times 1 + \frac{1}{3} \times \frac{3}{4} + \frac{1}{5} \times \frac{3}{4} + \frac{3}{10} \times \frac{2}{3} \\ &= \frac{1}{6} + \frac{1}{4} + \frac{3}{20} + \frac{1}{5} \\ &= \frac{10 + 15 + 9 + 12}{60} \\ &= \frac{46}{60} = \frac{23}{30} \end{aligned}$$

125. Let E_1, E_2 denote the events that the coin shows a head, tail and A be the event that the noted number is either 7 or 8.

We have,

$$P(E_1) = \frac{1}{2} \text{ and } P(E_2) = \frac{1}{2}$$

Now, $7 \rightarrow \{(1, 6), (6, 1), (2, 5), (5, 2), (3, 4), (4, 3)\}$

and $8 \rightarrow \{(2, 6), (6, 2), (3, 5), (5, 3), (4, 4)\}$

$$\text{Thus, } P(A/E_1) = \frac{11}{36}, P(A/E_2) = \frac{2}{11}$$

Hence, the required probability

$$\begin{aligned} P(A) &= P(E_1)P(A/E_1) + P(E_2)P(A/E_2) \\ &= \left(\frac{1}{2}\right)\left(\frac{11}{36}\right) + \left(\frac{1}{2}\right)\left(\frac{2}{11}\right) \\ &= \frac{193}{792} \end{aligned}$$

126. An urn contains m white and n black balls. A ball is drawn at random and is put into the urn along with k additional balls of the same color as that of the ball drawn. A ball is again drawn at random. What is the probability that the ball drawn now is white?

Baye's Theorem.

127. Hence, the required probability

$$\begin{aligned} &= \frac{\frac{1}{2} \times \frac{5}{11}}{\frac{1}{2} \times \frac{3}{7} + \frac{1}{2} \times \frac{5}{11}} \\ &= \frac{\frac{5}{11}}{\frac{3}{7} + \frac{5}{11}} \\ &= \frac{35}{33 + 35} = \frac{35}{68} \end{aligned}$$

128. Hence, the required probability

$$= \frac{\frac{1}{6} \times \frac{1}{100}}{\frac{1}{6} \times \frac{1}{100} + \frac{1}{3} \times \frac{3}{100} + \frac{1}{2} \times \frac{15}{100}}$$

$$= \frac{1}{1+6+45} = \frac{1}{52}$$

129. Hence, the required probability

$$= \frac{\frac{3}{5} \times \frac{1}{6}}{\frac{3}{5} \times \frac{1}{6} + \frac{2}{5} \times \frac{5}{6}}$$

$$= \frac{3}{3+5} = \frac{3}{8}$$

130. Hence, the required probability,

$$P(\overline{A}\overline{B} \text{ or } \overline{A}B) = P(\overline{A}\overline{B}) + P(\overline{A}B)$$

$$= P(A)P(\overline{B}) + P(\overline{A})P(B)$$

$$= \frac{3}{4} \times \frac{1}{6} + \frac{1}{4} \times \frac{5}{6}$$

$$= \frac{8}{24} = \frac{1}{3}$$

131. The probability of drawing a red ball = $1/6$

The probability of A speaks truth = $2/3$

The probability of B speaks truth = $4/5$

Thus, the probability that A and B both speak truth

$$= \frac{1}{6} \times \frac{2}{3} \times \frac{4}{5}$$

$$= \frac{4}{45}$$

Now, the probability that A falsely asserts a ball as red.

$$= \frac{1}{3} \times \frac{1}{5} = \frac{1}{15}$$

Similarly, the probability that B falsely asserts a ball as red.

$$= \frac{1}{5} \times \frac{1}{5} = \frac{1}{25}$$

The probability that both A and B speak false

$$= \frac{5}{6} \times \frac{1}{15} \times \frac{1}{25}$$

$$= \frac{1}{450}$$

Hence, the required probability

$$= \frac{\frac{4}{45}}{\frac{4}{45} + \frac{1}{450}}$$

$$= \frac{4}{45} = \frac{40}{41}$$

$$= \frac{40}{41}$$

132. Hence, the required probability

$$= \frac{\frac{1}{2} \times \frac{40}{100} \times \frac{90}{100}}{\frac{1}{2} \times \frac{60}{100} \times \frac{80}{100} + \frac{1}{2} \times \frac{40}{100} \times \frac{90}{100}}$$

$$= \frac{40 \times 90}{60 \times 80 + 40 \times 90}$$

$$= \frac{3600}{4800 + 3600}$$

$$= \frac{36}{48 + 36}$$

$$= \frac{36}{84} = \frac{3}{7}$$

133. Let E_1, E_2 and E_3 be the events that the answer is guessed, copied and knows the answer, respectively and E be the event that the examinee answers correctly.

$$\text{Given } P(E_1) = \frac{1}{3}, P(E_2) = \frac{1}{6}$$

Here, E_1, E_2 and E_3 are exhaustive events.

Thus,

$$P(E_1) + P(E_2) + P(E_3) = 1$$

$$\Rightarrow P(E_3) = 1 - P(E_1) - P(E_2)$$

$$= 1 - \frac{1}{3} - \frac{1}{6} = \frac{6-2-1}{6} = \frac{3}{6} = \frac{1}{2}$$

Now,

$P(E/E_1)$ = Probability of getting correct answer by guessing = $1/4$

$P(E/E_2)$ = Probability of getting correct answer by copying = $1/8$.

$P(E/E_3)$ = Probability of getting correct answer by knowing = 1 .

Hence, the required probability, $P(E_3/E)$

$$= \frac{P(E_3) \cdot P(E/E_3)}{P(E_1) \cdot P(E/E_1) + P(E_2) \cdot P(E/E_2) + P(E_3) \cdot P(E/E_3)}$$

$$= \frac{\frac{1}{2} \times 1}{\frac{1}{3} \times \frac{1}{4} + \frac{1}{6} \times \frac{1}{8} + \frac{1}{2} \times 1} = \frac{24}{29}$$

134. Let E_1/E_2 , denote the events that the black and red cards are lost and A denotes the event that the occurrence of 13 cards which are examined and found all red.

Obviously, $P(E_1) = \frac{1}{2} = P(E_2)$

$$\text{Also } P(A/E_1) = \frac{{}^{26}C_{13}}{{}^{51}C_{13}} \text{ and } P(A/E_2) = \frac{{}^{25}C_{13}}{{}^{51}C_{13}}$$

Hence the required probability,

$$\begin{aligned}
 P(E_1/A) &= \frac{P(E_1)P(A/E_1)}{P(E_1)P(A/E_1) + P(E_2)P(A/E_2)} \\
 &= \frac{\frac{1}{2} \cdot \frac{{}^{26}C_{13}}{{}^{51}C_{13}}}{\frac{1}{2} \cdot \frac{{}^{26}C_{13}}{{}^{51}C_{13}} + \frac{1}{2} \cdot \frac{{}^{25}C_{13}}{{}^{51}C_{13}}} \\
 &= \frac{{}^{26}C_{13}}{{}^{26}C_{13} + {}^{25}C_{13}} = \frac{2}{2+1} = \frac{2}{3}
 \end{aligned}$$

135. Let E be the event that an employee receive the letter and A be the event that employer received the reply.

$$\begin{aligned}
 P(E) &= \frac{n-1}{n} \text{ and } P(\bar{E}) = \frac{1}{n} \\
 P(A/E) &= \frac{n-1}{n} \text{ and } P(A/\bar{E}) = 0
 \end{aligned}$$

$$\begin{aligned}
 \text{Now, } P(A) &= P(E \cap A) + P(\bar{E} \cap A) \\
 &= P(E) \cdot P(A/E) + P(\bar{E}) \cdot P(A/\bar{E}) \\
 &= \left(\frac{n-1}{n}\right) \cdot \left(\frac{n-1}{n}\right) + \frac{1}{n} \cdot 0
 \end{aligned}$$

Hence the required probability,

$$\begin{aligned}
 P(E/\bar{A}) &= \frac{P(E \cap \bar{A})}{P(\bar{A})} \\
 &= \frac{P(E) - P(E \cap A)}{P(\bar{A})} \\
 &= \frac{P(E) - P(E) \cdot P(A/E)}{P(\bar{A})} \\
 &= \frac{\left(\frac{n-1}{n}\right) - \left(\frac{n-1}{n}\right) \cdot \left(\frac{n-1}{n}\right)}{\frac{2n-1}{n^2}} \\
 &= \frac{n-1}{2n-1}
 \end{aligned}$$

136. We have to find the probability of A being hit by B but not by C .

Thus,

$$\begin{aligned}
 P(BC'/A) &= \frac{P(BC') \cdot P(A/BC')}{P(BC') \cdot P(A/BC') + P(B'C) \cdot P(A/B'C) + P(BC) \cdot P(A/BC) + P(B'C') \cdot P(A/B'C')} \\
 &= \frac{1 \cdot \frac{1}{2} \cdot \frac{2}{3}}{1 \cdot \frac{1}{2} \cdot \frac{2}{3} + 1 \cdot \frac{1}{2} \cdot \frac{1}{3} + 1 \cdot \frac{1}{2} \cdot \frac{1}{3} + 0 \cdot \frac{1}{2} \cdot \frac{2}{3}} \\
 &= \frac{\frac{1}{3}}{\frac{1}{3} + \frac{1}{6} + \frac{1}{6}} = \frac{1}{2}
 \end{aligned}$$

137. Let E_1 be the event that the coin drawn is fair and E_2 be the event that the coin drawn is biased.

$$P(E_1) = \frac{m}{N} \text{ and } P(E_2) = \frac{N-m}{N}$$

Let A is the event that on tossing a coin the head appears first and then appears tail.

$$\begin{aligned}
 \text{Now, } P(A) &= P(E_1 \cap A) + P(E_2 \cap A) \\
 &= P(E_1) \cdot P(A/E_1) + P(E_2) \cdot P(A/E_2) \\
 &= \frac{m}{N} \left(\frac{1}{2}\right)^2 + \left(\frac{N-m}{N}\right) \left(\frac{2}{3}\right) \left(\frac{1}{3}\right)
 \end{aligned}$$

Hence the required probability

$$\begin{aligned}
 P\left(\frac{E_1}{A}\right) &= \frac{P(E_1 \cap A)}{P(A)} = \frac{P(E_1) \cdot P(A/E_1)}{P(A)} \\
 &= \frac{\frac{m}{N} \left(\frac{1}{2}\right)^2}{\frac{m}{N} \left(\frac{1}{2}\right)^2 + \frac{N-m}{N} \left(\frac{2}{3}\right) \left(\frac{1}{3}\right)} \\
 &= \frac{9m}{m+8N}
 \end{aligned}$$

138. Let E_i denotes the event that exactly i white balls are drawn.

$$\text{We have } P(E_i) = \frac{{}^6C_i \times {}^{12}C_{6-i}}{{}^{18}C_6}$$

Let E denotes the event that at least 4 white balls are drawn, then

$$E = E_4 \cup E_5 \cup E_6$$

Let A denotes the event that exactly one of the two balls next drawn is white.

$$\text{Now, } P(A/E_4) = \frac{{}^2C_1 \times {}^{11}C_1}{{}^{12}C_2} = \frac{20}{66},$$

$$\text{and } P(A/E_5) = \frac{{}^1C_1 \times {}^{11}C_1}{{}^{12}C_2} = \frac{1}{66}, P(A/E_6) = 0$$

Thus, the required probability,

$$\begin{aligned}
 P(A/E) &= \frac{P(A \cap E)}{P(E)} \\
 &= \frac{P(A \cap E_4) + P(A \cap E_5) + P(A \cap E_6)}{P(E)} \\
 &= \frac{P(E_4)P(A/E_4) + P(E_5)P(A/E_5) + P(E_6)P(A/E_6)}{P(E_4) + P(E_5) + P(E_6)} \\
 &= \frac{20 \times {}^6C_4 \times {}^{12}C_4 + 11 \times {}^6C_5 \times {}^{12}C_1}{66[{}^6C_4 \times {}^{12}C_2 + {}^6C_5 \times {}^{12}C_1 + 1]}
 \end{aligned}$$

139. Let C, S, B and T denote the events that the person goes to the office by a car, a scooter, a bus and a train respec-

tively and let L denote the event that he reaches late in the office.

We have,

$$P(C) = \frac{1}{7}, P(S) = \frac{2}{7}, P(B) = \frac{3}{7}, P(T) = \frac{1}{7}$$

$$P(L/C) = \frac{2}{9}, P(L/S) = \frac{4}{9}, P(L/B) = \frac{1}{9}, P(L/T) = \frac{1}{9}$$

$$\text{Thus, } P(L'/C) = \frac{7}{9}, P(L'/S) = \frac{5}{9},$$

$$P(L'/B) = \frac{8}{9}, P(L'/T) = \frac{8}{9}$$

Hence the required probability,

$$\begin{aligned} P(C/L') &= \frac{P(C)P(L'/C)}{P(C)P(L'/C) + P(S)P(L'/S) + P(B)P(L'/B) + P(T)P(L'/T)} \\ &= \frac{\frac{1}{7} \cdot \frac{7}{9}}{\frac{1}{7} \cdot \frac{7}{9} + \frac{2}{7} \cdot \frac{5}{9} + \frac{3}{7} \cdot \frac{8}{9} + \frac{1}{7} \cdot \frac{8}{9}} \\ &= \frac{1}{7} \end{aligned}$$

140. Let E_1, E_2, E_3, E_4 be the events of losing a card from spades, hearts, diamonds and clubs and E be the event of drawing 2 spades from the remaining cards.

Hence the required probability,

$$\begin{aligned} P(E_1/E) &= \frac{P(E_1)P(E/E_1)}{P(E_1)P(E/E_1) + P(E_2)P(E/E_2) + P(E_3)P(E/E_3) + P(E_4)P(E/E_4)} \\ &= \frac{\frac{{}^{13}C_1}{{}^{52}C_1} \times \frac{{}^{12}C_2}{{}^{51}C_2}}{\frac{{}^{13}C_1}{{}^{52}C_1} \times \frac{{}^{12}C_2}{{}^{51}C_2} + 3 \times \frac{{}^{13}C_1}{{}^{52}C_1} \times \frac{{}^{13}C_2}{{}^{51}C_2}} \\ &= \frac{\frac{1}{4} \times \frac{66}{1275}}{\frac{1}{4} \times \frac{66}{1275} + 3 \times \frac{1}{4} \times \frac{78}{1275}} \\ &= \frac{66}{66 + 78 + 78 + 78} \\ &= \frac{66}{300} = \frac{11}{50} \end{aligned}$$

141. Let T and L be the events that both A and B speak truth and lies and B the event that a ball is drawn from the bag.

$$\text{Clearly } P(B) = \frac{1}{6} \text{ and } P(\bar{B}) = \frac{5}{6}$$

$$P(T/B) = \frac{3}{5} \times \frac{7}{10}, P(L/\bar{B}) = \frac{2}{5} \times \frac{3}{10}$$

Hence, the required probability

$$\begin{aligned} &= \frac{\frac{1}{6} \times \frac{3}{5} \times \frac{7}{10}}{\frac{1}{6} \times \frac{3}{5} \times \frac{7}{10} + \frac{1}{6} \times \frac{2}{5} \times \frac{3}{10}} \\ &= \frac{21}{21+6} = \frac{21}{27} = \frac{7}{9} \end{aligned}$$

142. Let E_1, E_2 be the event that the letter came from CALCUTTA and TATANAGAR and A the event that two consecutive alphabets visible on the envelope are TA.

$$\text{Clearly, } P(E_1) = \frac{1}{2} = P(E_2)$$

$$\text{and } P(A/E_1) = \frac{1}{7}, P(A/E_2) = \frac{2}{8} = \frac{1}{4}$$

Hence the required probability,

$$\begin{aligned} P(E_2/A) &= \frac{P(E_2)P(A/E_2)}{P(E_1)P(A/E_1) + P(E_2)P(A/E_2)} \\ &= \frac{\frac{1}{2} \cdot \frac{1}{4}}{\frac{1}{2} \cdot \frac{1}{4} + \frac{1}{2} \cdot \frac{1}{4}} = \frac{4}{11} \end{aligned}$$

143. Let E_1, E_2 be the event that the letter came from MAHARASTRA and MADRAS and A the event that two consecutive alphabets visible on the envelope are RA

$$\text{Clearly, } P(E_1) = \frac{1}{2} = P(E_2)$$

$$\text{and } P(A/E_1) = \frac{2}{9}, P(A/E_2) = \frac{1}{5}$$

Hence the required probability,

$$\begin{aligned} P(E_2/A) &= \frac{P(E_2)P(A/E_2)}{P(E_1)P(A/E_1) + P(E_2)P(A/E_2)} \\ &= \frac{\frac{1}{2} \times \frac{2}{9}}{\frac{1}{2} \times \frac{2}{9} + \frac{1}{2} \times \frac{1}{5}} \\ &= \frac{\frac{2}{9}}{\frac{2}{9} + \frac{1}{5}} \\ &= \frac{10}{10+9} \\ &= \frac{10}{19} \end{aligned}$$

144. A can hit a target 4 times in 5 shots, B can hit 3 times in 4 shots and C twice in 3 shots. They fire once each. If two of them hit. What is the chance that C has missed it?

144.

145. Let E_1 be the event that both A and B speak truth, E_2 be the event that both A and B tell a lie, E be the event that A and B agree in a certain statement, C be the event that A speaks the truth and D be the event that B speaks truth.

$$\text{Thus, } E_1 = C \cap D, E_2 = C' \cap D'$$

$$\text{Therefore, } P(E_1) = P(C \cap D)$$

$$= P(C)P(D) = xy$$

$$\text{and } P(E_2) = P(C' \cap D')$$

$$= P(C')P(D')$$

$$= (1-x)(1-y)$$

$$= 1-x-y+xy$$

$$\text{Now, } P\left(\frac{E}{E_1}\right) = \text{Probability that } A \text{ and } B \text{ will agree}$$

when both of them speak the truth

$$= 1$$

$$\text{and } P\left(\frac{E}{E_2}\right) = \text{Probability that } A \text{ and } B \text{ will agree when}$$

both of them tell a lie

$$= 1$$

Hence, the required probability,

$$\begin{aligned} P\left(\frac{E}{E}\right) &= \frac{P(E_1) \cdot P\left(\frac{E}{E_1}\right)}{P(E_1) \cdot P\left(\frac{E}{E_1}\right) + P(E_2) \cdot P\left(\frac{E}{E_2}\right)} \\ &= \frac{xy \cdot 1}{xy \cdot 1 + (1-x-y+xy) \cdot 1} \\ &= \frac{xy}{1-x-y+2xy} \end{aligned}$$

$$146. P(E_1) = P(1 \text{ or } 3 \text{ on the die}) = \frac{2}{6} = \frac{1}{3}$$

$$P(E_2) = P(1 \text{ or } 3 \text{ not on the die}) = \frac{4}{6} = \frac{2}{3}$$

Let B be = the ball selected is black.

$$\text{Thus, } P\left(\frac{B}{E_1}\right) = \frac{3}{7}, P\left(\frac{B}{E_2}\right) = \frac{4}{7}$$

Therefore, the probability of a black ball

$$\begin{aligned} &= \frac{1}{3} \times \frac{3}{7} + \frac{2}{3} \times \frac{4}{7} \\ &= \frac{11}{21} \end{aligned}$$

Hence, the required probability

$$\begin{aligned} &= \frac{\frac{2}{3} \times \frac{4}{7}}{\frac{1}{3} \times \frac{3}{7} + \frac{2}{3} \times \frac{4}{7}} \\ &= \frac{8}{11} \end{aligned}$$

147. Given $S = \{HH, HT, TH, TT\}$

$$\text{and } R_x = \{0, 1, 2\}$$

$$\text{Now, } P(X=0) = \frac{1}{4}$$

$$P(X=1) = \frac{2}{4}$$

$$\text{and } P(X=2) = \frac{1}{4}$$

Hence, the probability distribution is

$$X: \quad 0 \quad 1 \quad 2$$

$$P(X): \quad \frac{1}{4} \quad \frac{2}{4} \quad \frac{1}{4}$$

148. Here, $R_x = \{0, 1, 2, 3, 4\}$

$$\text{Now, } P(X=0) = \frac{{}^6C_4}{{}^{10}C_4}$$

$$P(X=1) = \frac{{}^4C_1 \times {}^6C_3}{{}^{10}C_4}$$

$$P(X=2) = \frac{{}^4C_2 \times {}^6C_2}{{}^{10}C_4}$$

$$P(X=3) = \frac{{}^4C_3 \times {}^6C_1}{{}^{10}C_4}$$

$$\text{and } P(X=4) = \frac{{}^4C_4}{{}^{10}C_4}$$

Hence, the probability distribution is

$$X: \quad 0 \quad 1 \quad 2 \quad 3 \quad 4$$

$$P(X): \quad \frac{{}^6C_4}{{}^{10}C_4} \quad \frac{{}^4C_1 \times {}^6C_3}{{}^{10}C_4} \quad \frac{{}^4C_2 \times {}^6C_2}{{}^{10}C_4} \quad \frac{{}^4C_3 \times {}^6C_1}{{}^{10}C_4} \quad \frac{{}^4C_4}{{}^{10}C_4}$$

149. Here, $R_x = \{0, 1, 2\}$

$$\text{Now, } P(X=0) = \frac{5}{6} \times \frac{5}{6} = \frac{25}{36}$$

$$P(X=1) = \frac{1}{6} \times \frac{5}{6} + \frac{5}{6} \times \frac{1}{6} = \frac{10}{36}$$

$$\text{and } P(X=2) = \frac{1}{6} \times \frac{1}{6} = \frac{1}{36}$$

Hence, the required probability distribution

$$X: \quad 0 \quad 1 \quad 2$$

$$P(X): \quad \frac{25}{36} \quad \frac{10}{36} \quad \frac{1}{36}$$

150. Here, $R_x = \{0, 1, 2\}$

$$\text{Now, } P(X=0) = \frac{12}{13} \times \frac{12}{13} = \frac{144}{169}$$

$$P(X=1) = \frac{1}{13} \times \frac{12}{13} + \frac{12}{13} \times \frac{1}{13} = \frac{24}{169}$$

$$\text{and } P(X=2) = \frac{1}{13} \times \frac{1}{13} = \frac{1}{169}$$

Hence, the probability distribution is

$$\begin{array}{ccc} X: & 0 & 1 & 2 \\ P(X): & \frac{144}{169} & \frac{24}{169} & \frac{1}{169} \end{array}$$

151. Here, $R_x = \{0, 1, 2, 3\}$

$$\text{Now, } P(X=0) = \frac{{}^{48}C_3}{{}^{52}C_3}$$

$$P(X=1) = 3 \times \frac{{}^4C_1 \times {}^{48}C_2}{{}^{52}C_3}$$

$$P(X=2) = 3 \times \frac{{}^4C_2 \times {}^{48}C_1}{{}^{52}C_3}$$

$$\text{and } P(X=3) = \frac{{}^4C_3}{{}^{52}C_3}$$

Hence, the required probability distribution is

$$\begin{array}{cccc} X: & 0 & 1 & 2 & 3 \\ P(X): & \frac{{}^{48}C_3}{{}^{52}C_3} & \frac{3 \times {}^4C_1 \times {}^{48}C_2}{{}^{52}C_3} & \frac{3 \times {}^4C_2 \times {}^{48}C_1}{{}^{52}C_3} & \frac{{}^4C_3}{{}^{52}C_3} \end{array}$$

152. Here, $R_x = \{0, 1, 2, 3, 4\}$

$$\text{Now, } P(X=0) = \left(\frac{4}{5}\right)^4$$

$$P(X=1) = 4 \times \frac{1}{5} \times \left(\frac{4}{5}\right)^3$$

$$P(X=2) = 6 \times \left(\frac{1}{5}\right)^2 \times \left(\frac{4}{5}\right)^2$$

$$P(X=3) = 4 \times \left(\frac{1}{5}\right)^3 \times \left(\frac{4}{5}\right)$$

$$\text{and } P(X=4) = \left(\frac{1}{5}\right)^4$$

Hence, the required probability distribution is

$$\begin{array}{ccccc} X: & 0 & 1 & 2 & 3 & 4 \\ P(X): & \frac{256}{625} & \frac{256}{625} & \frac{96}{625} & \frac{16}{625} & \frac{1}{625} \end{array}$$

154. Let

$$2P(X=x_1) = 3P(X=x_2) = P(X=x_3) = 5P(X=x_4) = k$$

As we know that,

$$p_1 + p_2 + p_3 + p_4 = 1$$

$$\Rightarrow \frac{k}{2} + \frac{k}{3} + k + \frac{k}{5} = 1$$

$$\Rightarrow 15k + 10k + 30k + 12k = 30$$

$$\Rightarrow 67k = 30$$

$$\Rightarrow k = \frac{30}{67}$$

Hence, the required probability distribution is

$$\begin{array}{cccc} X: & x_1 & x_2 & x_3 & x_4 \\ P(X): & \frac{15}{67} & \frac{10}{67} & \frac{30}{67} & \frac{6}{67} \end{array}$$

155. Here $n=6$, $p=\frac{1}{6}$, $q=1-\frac{1}{6}=\frac{5}{6}$

$$\begin{aligned} \text{(i) } P(X \geq 5) &= P(X=5) + P(X=6) \\ &= {}^6C_5 p^5 q^1 + {}^6C_6 p^6 q^0 \end{aligned}$$

$$= 6p^5 q + p^6 = p^5(6p + q) = \left(\frac{1}{6}\right)^5 \times \left(\frac{11}{6}\right)$$

$$\begin{aligned} \text{(ii) } P(X \leq 5) &= 1 - P(X > 5) \\ &= 1 - P(X=6) \end{aligned}$$

$$= 1 - {}^6C_6 p^6 q^0 = 1 - \left(\frac{1}{6}\right)^6$$

$$\text{(iii) } P(X=5) = {}^6C_5 p^5 q^1 = 6 \times \left(\frac{1}{6}\right)^5 \times \left(\frac{5}{6}\right)$$

$$\text{(iv) } P(X=0) = {}^6C_0 p^0 q^6 = 6 \times \left(\frac{5}{6}\right)^6$$

156. Here, $n=6$, $p=\frac{1}{2}$, $q=1-\frac{1}{2}=\frac{1}{2}$

$$\text{(i) } P(X=5) = {}^6C_5 p^5 q^1 = 6 \times \left(\frac{1}{2}\right)^6$$

$$\begin{aligned} \text{(ii) } P(X \geq 5) &= P(X=5) + P(X=6) \\ &= {}^6C_5 p^5 q^1 + {}^6C_6 p^6 q^0 \end{aligned}$$

$$= 6 \times \left(\frac{1}{2}\right)^6 + \left(\frac{1}{2}\right)^6$$

$$= 7 \times \left(\frac{1}{2}\right)^6$$

$$\begin{aligned} \text{(iii) } P(X \leq 5) &= 1 - P(X > 5) \\ &= 1 - P(X=6) \end{aligned}$$

$$= 1 - \left(\frac{1}{2}\right)^6$$

$$\begin{aligned} \text{(iv) } P(X \geq 1) &= 1 - P(X < 1) \\ &= 1 - P(X=0) \end{aligned}$$

$$= 1 - \left(\frac{1}{2}\right)^6$$

$$\text{(v) } P(X=0) = \left(\frac{1}{2}\right)^6$$

157. Here, $n=7$, $p=\frac{1}{6}$, $q=\frac{5}{6}$

$$\text{(i) } P(X=0) = \left(\frac{5}{6}\right)^7$$

$$(ii) P(X=6) = {}^7C_6 \times \left(\frac{1}{6}\right)^6 \times \left(\frac{5}{6}\right)$$

$$(iii) P(X \geq 6) = P(X=6) + P(X=7) \\ = {}^7C_6 \times \left(\frac{1}{6}\right)^6 \times \left(\frac{5}{6}\right) + \left(\frac{1}{6}\right)^7$$

$$(iv) P(X \leq 6) = 1 - P(X > 6) \\ = 1 - P(X=7) \\ = 1 - \left(\frac{1}{6}\right)^7$$

158. Here $S = \{(1, 1), (1, 2), \dots, (6, 6)\}$ and $n = 5$.

Now, $7 \rightarrow \{(1, 6), (6, 1), (2, 5), (5, 2), (3, 4), (4, 3)\}$

$$\text{Thus, } p = \frac{6}{36} = \frac{1}{6}, q = 1 - \frac{1}{6} = \frac{5}{6}$$

Hence, the required probability,

$$P(X=4) = {}^7C_4 p^4 q^3 = 35 \times \left(\frac{1}{6}\right)^4 \times \left(\frac{5}{6}\right)^3$$

159. Here, $n = 5$, $p = \frac{1}{5}$, $q = 1 - \frac{1}{5} = \frac{4}{5}$.

Hence, the required probability,

$$P(X=4) = {}^5C_4 p^4 q^1 \\ = 5 \times \left(\frac{1}{5}\right)^4 \times \left(\frac{4}{5}\right) = \frac{4}{625}$$

160. Here, $n = 4$.

The possible outcome of doublet is $\{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6)\}$

$$\text{Thus, } p = \frac{6}{36} = \frac{1}{6}, q = 1 - \frac{1}{6} = \frac{5}{6}$$

Hence, the required probability,

$$P(X \geq 2) = 1 - P(X < 2) \\ = 1 - \{P(X=1) + P(X=0)\} \\ = 1 - \{{}^4C_1 p^1 q^3 + {}^4C_0 p^0 q^4\} \\ = 1 - (4pq^3 + q^4) \\ = 1 - q^3(4p + q) \\ = 1 - \frac{3}{2} \left(\frac{5}{6}\right)^3 \\ = 1 - \frac{125}{144} = \frac{19}{144}$$

161. Here, $p = \frac{1}{2}$, $q = 1 - \frac{1}{2} = \frac{1}{2}$

Hence, the required probability,

$$P(X=1 \text{ or } X=3 \text{ or } X=5 \text{ or } \dots \text{ to } \infty) \\ = P(X=1) + P(X=3) + P(X=5) + \dots \text{ to } \infty \\ = {}^nC_1 p^1 q^{n-1} + {}^nC_3 p^3 q^{n-3} + {}^nC_5 p^5 q^{n-5} + \dots$$

$$= {}^nC_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^{n-1} + {}^nC_3 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^{n-3} \\ + {}^nC_5 \left(\frac{1}{2}\right)^5 \left(\frac{1}{2}\right)^{n-5} + {}^nC_7 \left(\frac{1}{2}\right)^7 \left(\frac{1}{2}\right)^{n-7} + \dots \\ = \left(\frac{1}{2}\right)^n ({}^nC_1 + {}^nC_3 + {}^nC_5 + {}^nC_7 + \dots \text{ to } \infty) \\ = \left(\frac{1}{2}\right)^n \times 2^{n-1} = \frac{1}{2}$$

162. Here, $n = 10$, $p = 1 - \frac{1}{10} = \frac{9}{10}$, $q = \frac{1}{10}$

Hence, the required probability,

$$P(X=7) = {}^{10}C_7 p^7 q^3 \\ = {}^{10}C_7 \times \left(\frac{9}{10}\right)^7 \times \left(\frac{1}{10}\right)^3$$

163. Here, $n = 5$, $p = \frac{3}{4}$, $q = 1 - \frac{3}{4} = \frac{1}{4}$.

Hence, the required probability,

$$P(X \geq 3) = P(X=3) + P(X=4) + P(X=5) \\ = {}^5C_3 p^3 q^2 + {}^5C_4 p^4 q + {}^5C_5 p^5 q^0 \\ = 10p^3 q^2 + 5p^4 q + p^5 q^0 \\ = p^3(10q^2 + 5pq + p^2) \\ = \frac{27}{64} \left(\frac{10}{16} + \frac{15}{16} + \frac{3}{16}\right) \\ = \frac{27}{64} \times \frac{28}{64} \\ = \frac{27}{64} \times \frac{7}{16} = \frac{189}{1024}$$

164. Here, $n = 10$, $p = \frac{2}{7}$, $q = 1 - \frac{2}{7} = \frac{5}{7}$

Hence, the required probability,

$$P(X \geq 8) = P(X=8) + P(X=9) + P(X=10) \\ = {}^{10}C_8 p^8 q^2 + {}^{10}C_9 p^9 q^1 + {}^{10}C_{10} p^{10} q^0 \\ = p^8(45q^2 + 10pq + p^2) \\ = \left(\frac{2}{7}\right)^9 \times \left(\frac{45 \times 25}{49} + \frac{100}{49} + \frac{4}{49}\right) \\ = \left(\frac{2}{7}\right)^9 \times \left(\frac{1129}{49}\right)$$

165. Here, $n = 5$, $p = \frac{1}{2}$, $q = 1 - \frac{1}{2} = \frac{1}{2}$

Hence, the required probability,

$$P(X \geq 2) = 1 - P(X < 2) \\ = 1 - \{P(X=1) + P(X=0)\} \\ = 1 - \{{}^5C_1 p^1 q^4 + {}^5C_0 p^0 q^5\}$$

$$= 1 - \left(\frac{5}{32} + \frac{1}{32} \right)$$

$$= 1 - \frac{6}{32} = \frac{26}{32} = \frac{13}{16}$$

166. Hence, the required probability,

$$P(X=r) = {}^nC_r p^r q^{n-r}$$

$$= {}^nC_r \times \left(\frac{1}{m} \right)^r \times \left(\frac{m-1}{m} \right)^{n-r}$$

$$= {}^nC_r \times \frac{(m-1)^{n-r}}{m^n}$$

167. Hence, the required probability,

$$P(X=5 \text{ or } X=6) = P(X=5) + P(X=6)$$

$$= {}^{11}C_5 p^5 q^6 + {}^{11}C_6 p^6 q^5$$

$$= {}^{11}C_5 p^5 q^6 + {}^{11}C_5 p^6 q^5$$

$$= {}^{11}C_5 p^5 q^5 (p+q)$$

$$= {}^{11}C_5 p^5 q^5$$

$$= {}^{11}C_5 \times \left(\frac{2}{5} \right)^5 \times \left(\frac{3}{5} \right)^5$$

168. It is given that

$$P(X \geq 1) > \frac{2}{3}$$

$$\Rightarrow 1 - P(X < 1) > \frac{2}{3}$$

$$\Rightarrow 1 - P(X=0) > \frac{2}{3}$$

$$\Rightarrow 1 - {}^nC_0 p^0 q^n > \frac{2}{3}$$

$$\Rightarrow 1 - q^n > \frac{2}{3}$$

$$\Rightarrow q^n < 1 - \frac{2}{3}$$

$$\Rightarrow \left(\frac{3}{4} \right)^n < \frac{1}{3}$$

Clearly, $n = 4$.

170. Probability that any one thing is received by a man

$$= \frac{a}{a+b} = p(\text{say})$$

Probability that any one thing is received by a woman

$$= \frac{b}{a+b} = q(\text{say})$$

Clearly, $p+q=1$

Out of m things, if r received by men, the rest $(m-r)$ will be received by women.

Hence, the required probability,

$$P(1) + P(3) + P(5) + \dots$$

$$= {}^mC_1 p q^{m-1} + {}^mC_3 p^3 q^{m-3} + {}^mC_5 p^5 q^{m-5} + \dots$$

$$= \frac{1}{2} [(p+q)^m - (p-q)^m]$$

$$= \frac{1}{2} [1 - (p-q)^m]$$

$$= \frac{1}{2} \left[1 - \left(\frac{b-a}{a+b} \right)^m \right]$$

$$= \frac{1}{2} \left[\frac{(b+a)^m - (b-a)^m}{(a+b)^m} \right]$$

171. Let X be the number of games A wins against B and p be the probability A wins the game against B .

Now,

$$P(A \text{ wins the match}) = P(X \geq 2)$$

$$= P(X=2) + P(X=3)$$

$$= {}^3C_2 p^2 q + {}^3C_3 p^3$$

$$= 3p^2 q + p^3$$

$$= 3 \cdot (0.4)^2 \cdot (0.6) + (0.4)^3$$

$$= (0.4)^2 (3 \cdot (0.6) + (0.4))$$

$$= (0.16) \cdot (2.2) = 0.352$$

Also,

$$P(A \text{ wins the match}) = P(X \geq 3)$$

$$= P(X=3) + P(X=4) + P(X=5)$$

$$= {}^5C_3 p^3 q^2 + {}^5C_4 p^4 q + {}^5C_5 p^5$$

$$= 10p^3 q^2 + 5p^4 q + p^5$$

$$= p^3 (10q^2 + 5pq + p^2)$$

$$= (0.4)^3 (10(0.6)^2 + 5(0.24) + (0.4)^2)$$

$$= (0.4)^3 (3.6 + 1.2 + .16)$$

$$= (0.064)(4.96)$$

$$= 0.31744$$

It shows that, best of 3 games 'option is better'.

172. Here $7 \rightarrow (1, 6), (6, 1), (2, 5), (5, 2), (3, 4), (4, 3)$

Now, the probability of getting 7 = $p = \frac{1}{6}$

and the probability of not getting 7 = $q = \frac{5}{6}$

It is given that,

$$P(X \geq 1) > \frac{95}{100}$$

$$\Rightarrow 1 - P(X < 1) > \frac{19}{20}$$

$$\Rightarrow 1 - P(X=0) > \frac{19}{20}$$

$$\Rightarrow 1 - {}^nC_0 p^0 q^n > \frac{19}{20}$$

$$\Rightarrow 1 - q^n > \frac{19}{20}$$

$$\Rightarrow 1 - \left(\frac{5}{6} \right)^n > \frac{19}{20}$$

$$\Rightarrow \left(\frac{5}{6} \right)^n < 1 - \frac{19}{20}$$

$$\Rightarrow \left(\frac{5}{6}\right)^n < \frac{1}{20}$$

Clearly, n is 4.

173. Let X be the event of getting 1, 3 or 5 in a single throw of a die.

$$P(X) = \frac{3}{6} = \frac{1}{2}$$

Hence, the required probability,

$$P(X=0) + P(X=1) + P(X=3) + \dots$$

$$\begin{aligned} &= {}^{2n+1}C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^{2n+1} + {}^{2n+1}C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^{2n} \\ &\quad + \dots + {}^{2n+1}C_n \left(\frac{1}{2}\right)^n \left(\frac{1}{2}\right)^{n+1} \\ &= \frac{1}{2^{2n+1}} ({}^{2n+1}C_0 + {}^{2n+1}C_1 + {}^{2n+1}C_2 + \dots + {}^{2n+1}C_n) \\ &= \frac{2^{2n}}{2^{2n+1}} \\ &= \frac{1}{2} \end{aligned}$$

174. Here, $n = 10$, $p = 5/6$ and $q = 1/6$

Hence, the required probability,

$$\begin{aligned} P(X < 2) &= P(X=0) + P(X=1) \\ &= {}^{10}C_0 p^0 q^{10} + {}^{10}C_1 p^1 q^9 \\ &= q^{10} + 10p^1 q^9 \\ &= q^9 (q + 10p) \\ &= \left(\frac{1}{6}\right)^9 \left(\frac{1}{6} + \frac{50}{6}\right) \\ &= \left(\frac{1}{6}\right)^9 \times \frac{51}{6} \\ &= \left(\frac{1}{6}\right)^9 \times \frac{17}{2} \\ &= \frac{17}{2 \times 6^9} \end{aligned}$$

175. Here $n = 4$, $p = 1/3$ and $q = 2/3$

Now, mean = $np = 4/3$

Variance = $npq = 8/9$

Standard Deviation = $+\sqrt{\text{Var}(X)}$

$$\begin{aligned} &= +\sqrt{\frac{8}{9}} \\ &= \frac{2\sqrt{2}}{3} \end{aligned}$$

176. Here, $np = 4$, $npq = 4/3$

Thus, $p = 2/3$, $q = 1/3$, $n = 6$

Hence, the required probability = $P(X \geq 1)$

$$\begin{aligned} &= 1 - P(X < 1) \\ &= 1 - P(X=0) \\ &= 1 - {}^{10}C_0 p^0 q^6 \end{aligned}$$

$$\begin{aligned} &= 1 - \left(\frac{1}{3}\right)^6 \\ &= \left(\frac{3^6 - 1}{3^6}\right) \end{aligned}$$

177. Here, $n = 5$

It is given that,

$$\begin{aligned} np + npq &= 1.8 \\ \Rightarrow np(1 + p) &= 1.8 \\ \Rightarrow 5p(1 + 1 - p) &= \frac{9}{5} \\ \Rightarrow 25p(2 - p) &= 9 \\ \Rightarrow 50p - 25p^2 - 9 &= 0 \\ \Rightarrow -50p + 25p^2 + 9 &= 0 \\ \Rightarrow 25p^2 - 50p + 9 &= 0 \\ \Rightarrow 25p^2 - 5p - 45p + 9 &= 0 \\ \Rightarrow 5p(5p - 1) - 9(5p - 1) &= 0 \\ \Rightarrow (5p - 9)(5p - 1) &= 0 \\ \Rightarrow p &= \frac{1}{5}, \frac{9}{5} \\ \Rightarrow p &= \frac{1}{5} \end{aligned}$$

Thus, $q = \frac{4}{5}$

Hence, the binomial distribution,

$$\begin{aligned} P(X=r) &= {}^nC_r p^r q^{n-r} \\ &= {}^6C_r \left(\frac{1}{5}\right)^r \left(\frac{4}{5}\right)^{6-r} \end{aligned}$$

where $r = 0, 1, 2, 3, 4, 5, 6$.

178. Let A be the event in the call taking place in the last 20 minutes of the hour.

Let us represent the sample space as a line segment of length 20.

Hence the required probability

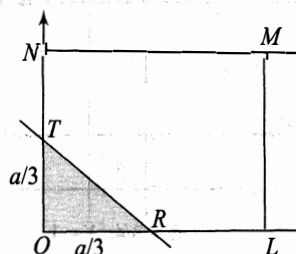
$$\begin{aligned} &= \frac{\text{Measure of specified part of the region}}{\text{Measure of the whole region}} \\ &= \frac{20}{60} = \frac{1}{3} \end{aligned}$$

179. Consider two points P and Q on the segments CB and CA respectively.



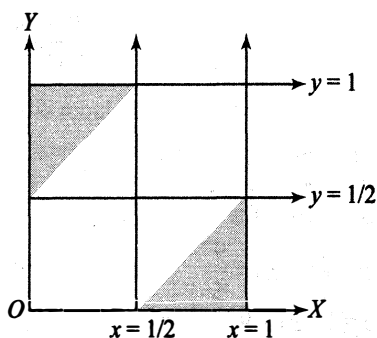
Let $CP = y$ and $CQ = x$ so that

$$0 \leq x \leq a/2, 0 \leq y \leq a/2$$



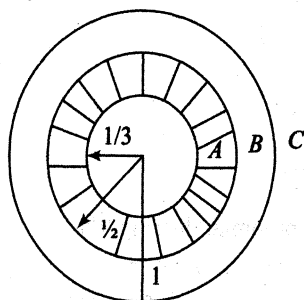
We are interested in the event $x + y < a/3$.

Here, first we make a square of side $a/2$ and its sub-region triangle with base $a/3$ and height is also $a/3$.



$$\text{Thus, the required probability} = \frac{\frac{1}{2} \left(\frac{a}{3} \right) \left(\frac{a}{3} \right)}{\left(\frac{a}{2} \right) \left(\frac{a}{2} \right)} = \frac{2}{9}$$

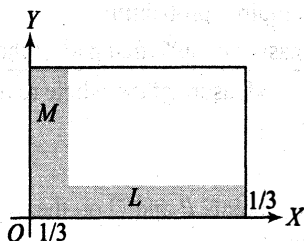
180. Let the two numbers be x and y such that $0 \leq x, y \leq 1$.



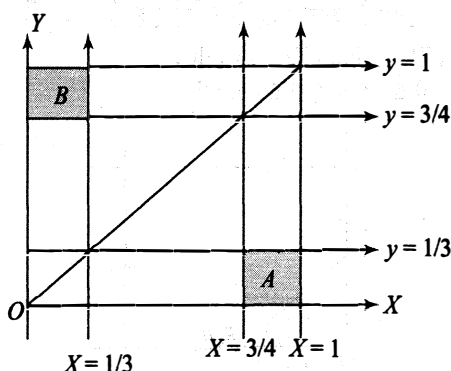
Now, the probability of

$$P\left(|x-y| > \frac{1}{2}\right) = \frac{\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} + \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}}{1 \times 1} = \frac{1}{4}$$

181. Let us consider two numbers be x and y where $0 \leq x, y \leq 1$, so that the geometric configuration is a square.



Let $M = \max. \{x, y\}$ and $m = \min. \{x, y\}$



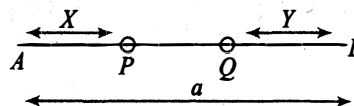
Hence the required probability,

$$P\left(\frac{M \geq 3/4}{m \leq 1/3}\right) = \frac{P(M \geq 3/4, m \leq 1/3)}{P(m \leq 1/3)}$$

$$= \frac{2 \cdot \frac{1}{3} \cdot \frac{1}{4}}{\frac{1}{3} + \frac{1}{3} - \frac{1}{9}} = \frac{\frac{2}{12}}{\frac{5}{9}} = \frac{3}{10}$$

182. Let AB be the straight line divided at P and Q such that $AB = a$.

Let $AP = x$ and $BQ = y$



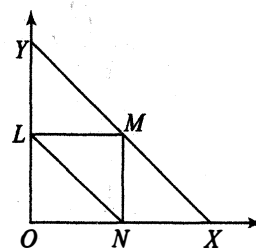
Now in favourable cases, we must have,

$$x < a/2, y < a/2$$

and $x + y > a/2$

Hence, the required probability

$$= \frac{\frac{1}{2} \cdot \frac{a}{2} \cdot \frac{a}{2}}{\frac{1}{2} \cdot a \cdot a} = \frac{1}{4}$$



- 183 Let two numbers be x and y such that $x + y = 24$.

For extremum, it is clear that, the greatest value of xy is $(12)^2$

Hence the required probability,

$$P(xy \geq 108) = P(x(24-x) \geq 108)$$

$$= P((x-18)(x-6) \leq 0)$$

$$= P(6 \leq x \leq 18)$$

$$= \frac{12}{24} = \frac{1}{2}$$

184. Let the probability of a shot falling in the region C of radius $1/3$ is

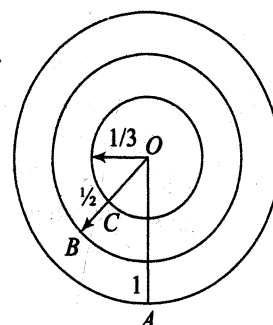
$$P_1 = \frac{\pi \cdot (1/3)^2}{\pi \cdot (1)^2} = \frac{1}{9}$$

the probability of a shot falling in the region B of radius $1/2$ is

$$P_2 = \frac{\pi \cdot (1/2)^2 - \pi \cdot (1/3)^2}{\pi \cdot (1)^2} = \frac{5}{36}$$

and the probability of a shot falling in the region A of radius 1 is

$$P_3 = \frac{\pi \cdot (1)^2 - \pi \cdot (1/2)^2}{\pi \cdot (1)^2} = \frac{3}{4}$$



Hence the required probability is

$$= P(p_1 \cdot p_2 \cdot p_3 \times 3!) = \frac{6 \times 5}{432} = \frac{5}{72}.$$

185. Let A be the event that the length of the interval in which the given equation has real roots and S be the sample space, i.e. the length of the interval $[0, 5]$. Since roots are real,

$$D \geq 0 \Rightarrow p^2 - 4 \cdot 1 \cdot \frac{(p+2)}{4} \geq 0$$

$$\Rightarrow p^2 - p - 2 \geq 0$$

$$\Rightarrow (p-2)(p+1) \geq 0$$

$$\Rightarrow p \leq -1, p \geq 2$$

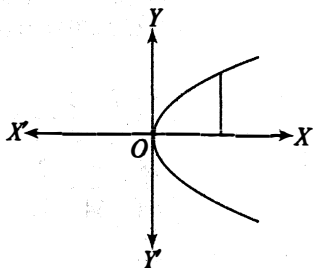
Thus, $A = [2, 5]$, since $p \in [0, 5]$.

$$\text{Hence the required probability} = \frac{n(A)}{n(S)} = \frac{3}{5}.$$

186. Consider a point with co-ordinates (x, y) corresponds to the pair of numbers x and y .

The sample space is a square whose sides are unit segments to the co-ordinate axes.

The region, whose set of points corresponds to the outcome favourable to the event $y^2 \leq x$ is bounded by the graphs of the function $y = 0$, $x = 1$ and $y^2 = x$.



Hence the required

$$\text{probability} = \int_0^1 \sqrt{x} \, dx = \left(\frac{2}{3} x^{3/2} \right)_0^1 = \frac{2}{3}.$$

187. Let A and B arrive at the place of their meeting x and y mins. after 11 am.

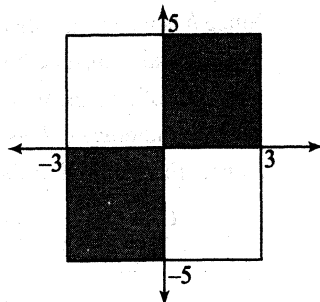
As per the condition, their meeting is possible only when $|x - y| \leq 20$.

Hence the required probability

$$\begin{aligned} &= \frac{60 \times 60 - \frac{1}{2} \times 40 \times 40 - \frac{1}{2} \times 40 \times 40}{60 \times 60} \\ &= \frac{2000}{3600} = \frac{5}{9} \end{aligned}$$

188. Hence, the required probability

$$\begin{aligned} &= \frac{15+15}{6 \times 10} \\ &= \frac{30}{60} = \frac{1}{2} \end{aligned}$$



189. Clearly, $x \in [-2, 3]$

Now, the solution set of $\frac{(x+1)(x-2)}{(x-4)} \geq 0$ is $x \in [-1, 2]$.

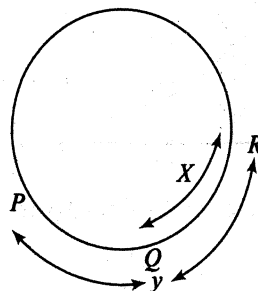
Hence, the required probability = $\frac{3}{5}$.

190. Let the length of the circumference is $2s$.

Let x denotes the clockwise arc length of PQ and let y denotes the clockwise arc length of PR .

Thus, $0 < x < 2s, 0 < y < 2s$

...(i)

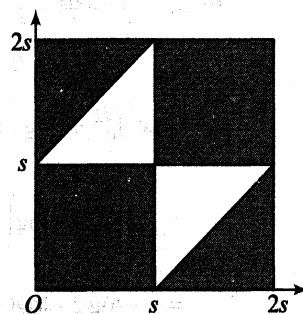


Let A denotes the subset of s for which any of the following conditions hold.

- (i) $x, y < s$
- (ii) $x < s$ and $y - x > s$
- (iii) $x, y > s$
- (iv) $y < s$ and $x - y > s$

Thus A consists of these points for which P, Q , and R lie on a semi-circle.

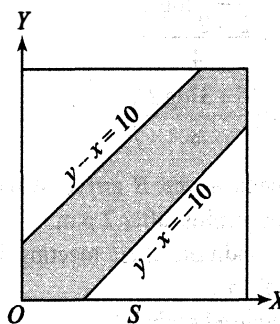
$$\text{Thus, } p = \frac{\text{Area of } A}{\text{Area of } S} = \frac{3s^2}{4s^2} = \frac{3}{4}.$$



191. The sample space is

$$S = \{(x, y) : 0 \leq x, y \leq 30\}$$

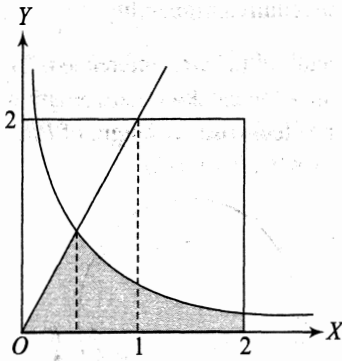
$$\text{Let } A = \{(x, y) : |x - y| \leq 10\}$$



Hence, the required probability

$$\begin{aligned} &= \frac{30 \times 30 - \frac{1}{2} \times 20 \times 20 - \frac{1}{2} \times 20 \times 20}{30 \times 30} \\ &= \frac{900 - 400}{900} \\ &= \frac{500}{900} \\ &= \frac{5}{9} \end{aligned}$$

193. Clearly, $0 < x, y \leq 2$.



Let A be the area of the shaded part

$$\begin{aligned}
 \text{Now, } A &= \int_0^{\frac{1}{\sqrt{2}}} 2x \, dx + \int_{\frac{1}{\sqrt{2}}}^2 \frac{dx}{x} \\
 &= (x^2)_0^{\frac{1}{\sqrt{2}}} + (\log |x|)_1^{\frac{1}{\sqrt{2}}} \\
 &= \frac{1}{2} + \log 2 - \log \left(\frac{1}{\sqrt{2}} \right) \\
 &= \frac{1}{2} + \log 2 + \log(\sqrt{2}) \\
 &= \frac{1}{2} + \log 2 + \frac{1}{2} \log(2) \\
 &= \frac{1}{2} + \frac{3}{2} \log 2
 \end{aligned}$$

Hence, the required probability,

$$\begin{aligned}
 P(A) &= \frac{\frac{1}{2} + \frac{3}{2} \log 2}{4} \\
 &= \left(\frac{1 + 3 \log 2}{8} \right)
 \end{aligned}$$

194. Let two friends A and B arrive at the place of their meeting x and y mins. after 2 p.m.

As per the condition, their meeting is possible only when $|x - y| \leq 15$.

Hence the required probability

$$\begin{aligned}
 &= \frac{60 \times 60 - \frac{1}{2} \times 45 \times 45 - \frac{1}{2} \times 45 \times 45}{60 \times 60} \\
 &= \frac{3600 - 2025}{3600} \\
 &= \frac{1575}{3600} \\
 &= \frac{63}{144} \\
 &= \frac{7}{16}
 \end{aligned}$$